



AQR Alternative Thinking 2026 Issue 3

Inflation Redux?

Real Solutions for Real Returns

Portfolio Solutions Group

Global inflation rates fell quickly after their 2022 peak, but inflation uncertainty lingers in several major markets including the U.S. At the time of writing in September 2026, equity markets appear unconcerned, but bond markets are more circumspect. How should investors be thinking about inflation risk? Can portfolios be made more resilient without making a directional bet on the future path of inflation? And how should investors evaluate the most inflation-resilient strategies?

This article updates AQR's earlier analysis on inflation sensitivities and inflation protection (published just before the 2022 shock), highlighting the benefits of commodities and trend following strategies and how to evaluate different implementations. It asks, "How should you prepare for the possibility of another global inflation shock?"

Contents

Introduction: How Should Investors Think About Inflation Risk?	3
Measuring Inflation Sensitivity	4
Reviewing Inflation Sensitivities of Major Asset Classes	6
Inflation-Aware Asset Allocation	9
The Stock/Bond Correlation	9
Inflationary and Disinflationary Tail Risks	10
Robust Asset Allocation Case Study	11
Implementation: Optimizing Resilience/Return Trade-Offs	13
Commodities	14
Trend Following	15
Concluding Thoughts	16
References and Further Reading	17
Appendix	18

About the Portfolio Solutions Group

The Portfolio Solutions Group (PSG) provides thought leadership to the broader investment community and custom analyses to help AQR clients achieve better portfolio outcomes.

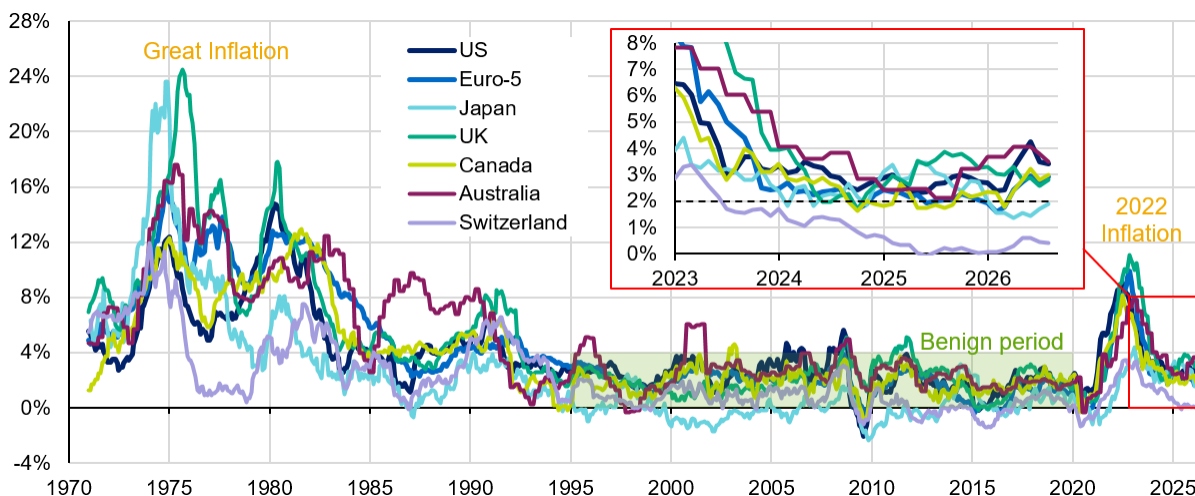
Authors Pete Hecht, Antti Ilmanen, Thomas Maloney and Nick McQuinn thank colleagues Alfie Brixton, Jordan Brooks, Ed Rayment and Erik Stamelos for helpful comments.

Introduction: How Should Investors Think About Inflation Risk?

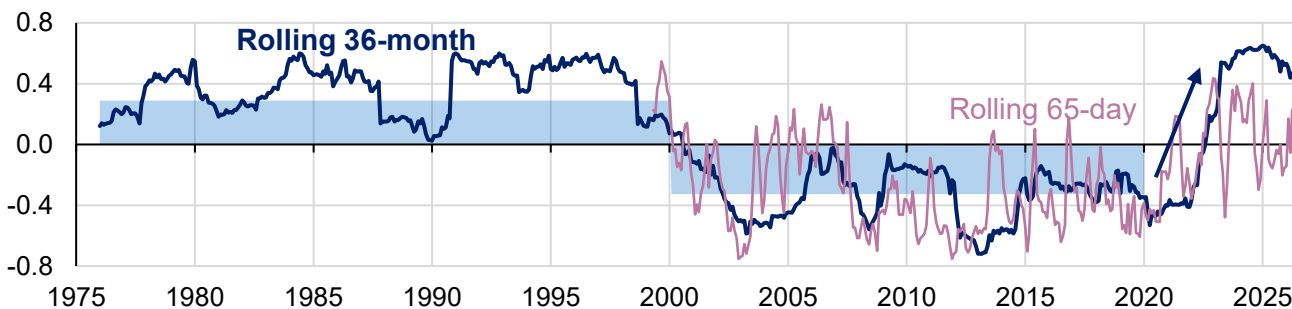
The quarter century from the mid-1990s to 2020 was a period of low and stable inflation across developed markets, well-anchored inflation expectations, and credible central banks facing few trade-offs. This era ended abruptly in the early 2020s, with a wave of inflation that caused a brief bear market in equities and a longer-lasting repricing of bond markets. Since then, most inflation rates have fallen, but in some major economies (notably the U.S. and U.K.) they have remained stubbornly higher than central bank targets (red box in **Exhibit 1** Chart A). Japan continues to experience its first significant bout of inflation in over 30 years. Public debt levels are high and rising in several major economies, the Fed's independence has been threatened by the current U.S. administration, trade policy has been volatile, and geopolitical events in 2026 have added to existing inflation pressures and uncertainty. It is hardly surprising that stock/bond correlations – which depend mainly on the relative importance of inflation news versus growth news¹ – have recently risen to multi-decade highs (Chart B). We discuss the implications of this for asset allocation in a later section.

Exhibit 1: Knowing Your Inflation History

A. Year-on-Year CPI Inflation Rates in Major Markets January 1, 1970 – July 31, 2026



B. Rolling Correlation Between U.S. Stocks and U.S. Treasuries January 1, 1973 – July 31, 2026



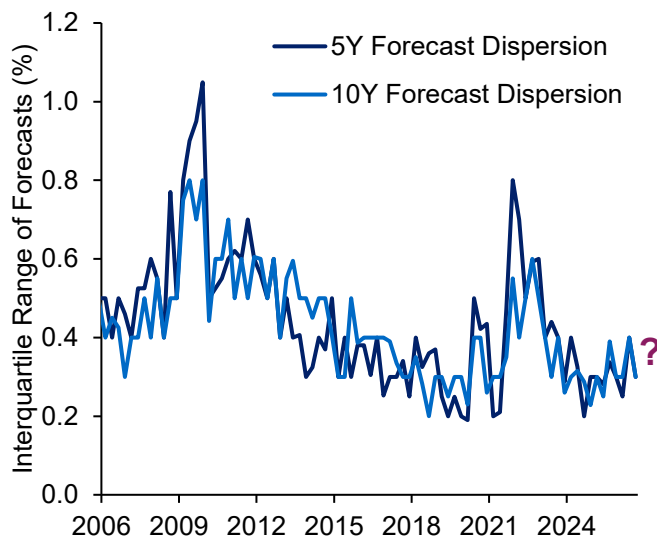
Sources: Bloomberg and AQR. Upper chart based on monthly data. Euro-5 is simple average of France, Germany, Italy, Netherlands and Spain. Lower chart shows correlations between MSCI USA and Bloomberg U.S. Treasury Index excess-of-cash returns, where cash is BoA ML 3-month T-Bill Index. Blue shaded areas represent full period correlation within the two regime sub-samples (1973-1999 and 2000-2019).

¹ Growth news tends to push stock and bond markets in opposite directions, while inflation news tends to push them in the same direction. For detailed analysis and discussion, see Brixton et al. (2023) "A Changing Stock-Bond Correlation: Drivers and Implications".

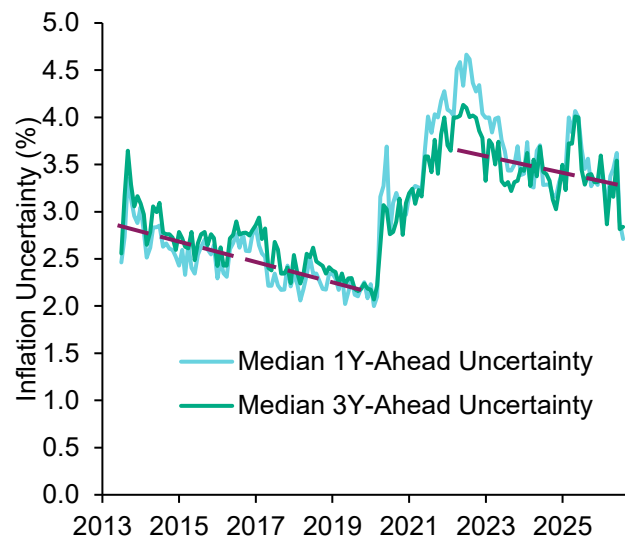
And yet, so far, longer-term inflation expectations have remained well-anchored throughout the 2020s turbulence. Buoyant equity market valuations at the time of writing reflect strong corporate earnings and appear untroubled by macroeconomic uncertainties. New technologies tend to be disinflationary, and AI may fit this pattern. Economists' forecasts became dispersed in 2022 but returned to close agreement by 2024, seeing only a small upward drift in 2025-26 (**Exhibit 2** Chart A). In contrast, consumer uncertainty has remained elevated in the years following the 2022 inflation shock (Chart B).

Exhibit 2: Tracking Inflation Uncertainty with Survey Data

A. Dispersion in Economists' Forecasts of Long-Term Inflation, January 2006 - July 2026



B. Consumer Inflation Uncertainty June 2013 - July 2026



Sources: Federal Reserve Banks of Philadelphia and New York. Left chart shows dispersion among participants in the Survey of Professional Forecasters. Right chart shows an estimated interquartile range based on probabilities assigned by participants to different inflation outcomes, from the Survey of Consumer Expectations.

Should investors still be worried about inflation? Well, a new inflation shock is only one of several tail risks they should consider, but it is particularly serious for any equity-dominated portfolio that relies solely on bonds for macro diversification, as we'll see. And we'll show that there may be some prudent steps investors can take that won't cost the earth if a "second wave" of inflation doesn't materialize after all.

Inflation tail risks come in different flavors: global supply shock, overheating economy, stagflation. In this article we focus our analysis on the expected impact of a 12-month inflation shock on investor portfolios, and how to mitigate it. One of the most serious tail risks is a de-anchoring of longer-term inflation expectations, last seen in major economies in the 1970s, which could have a long-lasting impact on markets. We discuss this in our conclusion.

Measuring Inflation Sensitivity

There are many challenges when measuring the sensitivity of asset prices to inflation. Asset prices likely already reflect prevailing levels of inflation and instead respond to inflation news. Therefore, when measuring sensitivities, it is critical to extract the news (or unexpected) component of inflation. In their 2021

paper “When Stock-Bond Diversification Fails: Managing Inflation Risk in Investor Portfolios,” the AQR authors combined two measures of the “information content” of realized U.S. inflation:

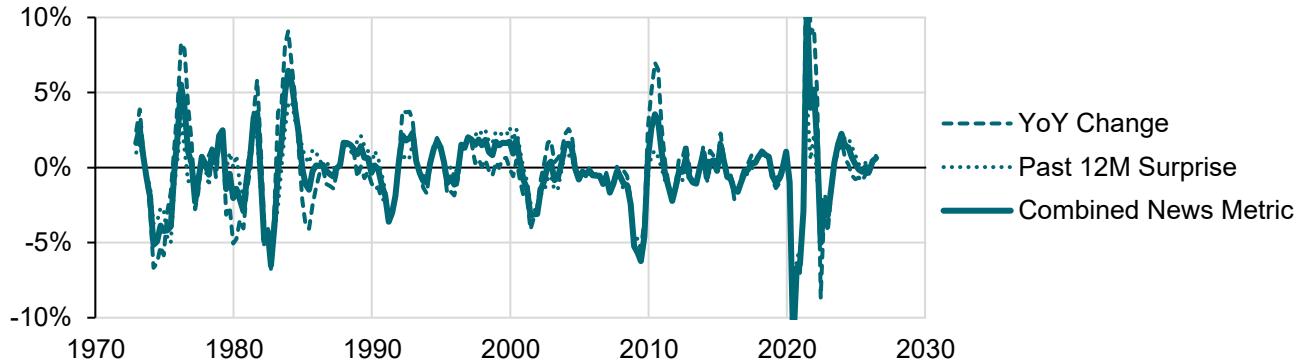
- 1) year-on-year CPI inflation minus CPI for previous 1-year period (“change”)
- 2) year-on-year CPI inflation minus 1-year forecast at start of period (“surprise”)

Both can be thought of as measures of the surprise in inflation – with the first using a “random walk” model for the path of expectations and the second using surveys. While neither is perfect, the combination may serve to reduce noise in either one. By focusing on quarterly overlapping year-on-year periods, the metric avoids seasonal effects and mitigates the role of publication lags. We follow the same methodology, and, as before, we construct a corresponding metric for U.S. GDP growth, as a control variable. We’ll also explore sensitivities to the main components of headline inflation – core, energy and food.

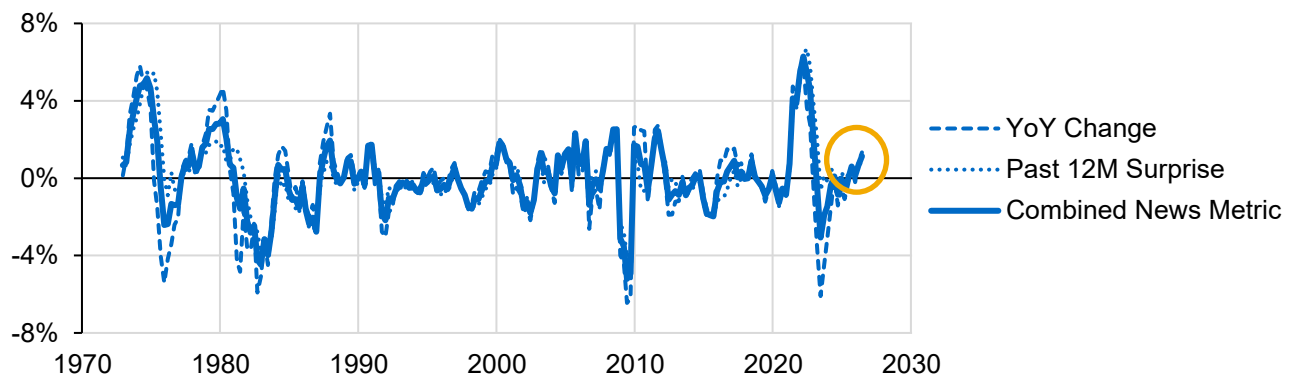
The choice of sample period is important: there is a trade-off between including more data versus focusing on more recent periods with greater relevance and better data quality and availability. As a middle ground, we choose to analyze the 55-year period from 1972 to 2026, which is largely relevant to the current regime, allows us to test a wide range of assets and strategies, and still encompasses a wide range of different inflation environments as shown in **Exhibit 3**. The latest uptick in inflation news is highlighted in Chart B. Our analysis focuses on U.S. inflation, where the best data are available.

Exhibit 3: U.S. Growth and Inflation News Indicators January 1, 1972 – June 30, 2026

A. Growth News Indicator



B. Inflation News Indicator



Source: Fed Survey of Professional Forecasters, U.S. Bureau of Labor Statistics, Bloomberg and AQR. Data at quarterly frequency. Changes calculated as difference between year-on-year growth or inflation and year-on-year growth or inflation 12 months earlier. Surprises calculated as difference between year-on-year growth or inflation and 1-year forecast a year earlier. Please read important disclosures in the Appendix.

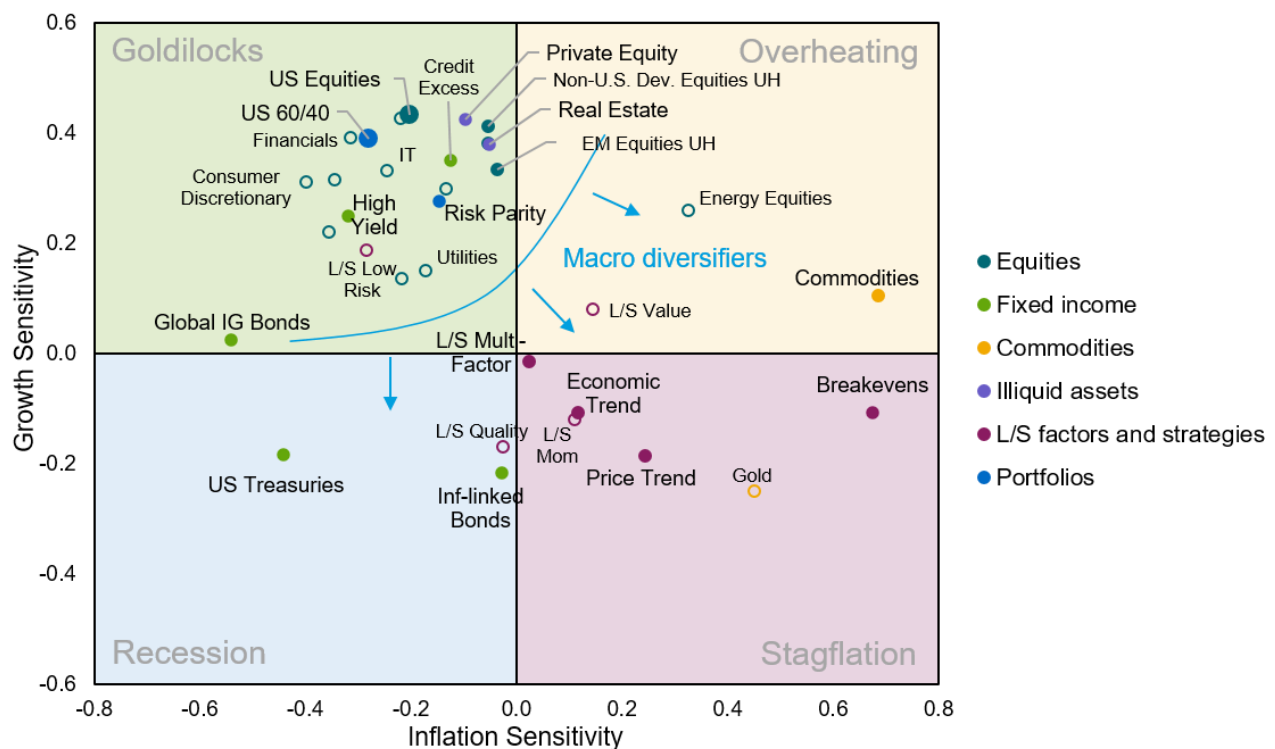
If we limit ourselves to more recent periods, we can measure inflation news at much higher frequencies, using inflation swaps, breakevens, CPI nowcast revisions, and analysis of inflation sentiment in text data, and focus on market responses around announcements. These are critical inputs for managers seeking to position dynamically to harness economic trends, as we'll discuss later. But for investors seeking to understand and mitigate the biggest risks to their portfolios, we think it makes sense to focus on longer data sets and longer horizons: a prolonged inflation shock will be most consequential to real wealth accumulation.

Reviewing Inflation Sensitivities of Major Asset Classes

Exhibit 4 uses the above metrics to map the average growth and inflation sensitivities of a range of asset classes, sectors and strategies over more than half a century. Broad equity markets and most equity sectors are situated firmly in the top left “Goldilocks” quadrant, meaning higher returns are associated with upside growth news and downside inflation news. Most investor portfolios are also in this quadrant, which means they have an implied bet on positive growth and low or falling inflation. It is difficult to avoid these sensitivities – after all, expected returns are, in part, compensation for bearing these risks. But any investment with very different macroeconomic sensitivities (those below and right of the blue line) will help to shift the portfolio towards the center, making performance more resilient across different environments.

Exhibit 4: The Long View: Growth and Inflation Sensitivities of Assets and Hypothetical Strategies

January 1, 1972 – June 30, 2026

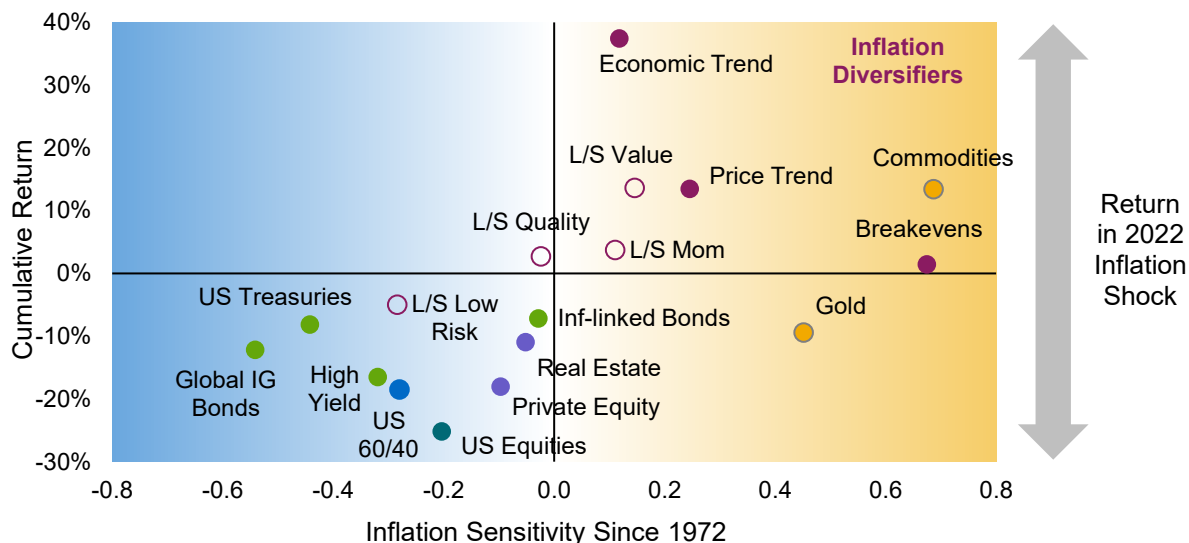


Source: AQR, Bloomberg, Survey of Professional Forecasters, U.S. Bureau of Labor Statistics. Sensitivities are two-factor partial contemporaneous correlations to U.S. inflation and growth news metrics, where each metric is an equal-risk combination of 12-month changes and 12-month surprises vs. survey forecasts. Based on rolling 12-month periods at quarterly frequency. Unfilled markers indicate sub-sectors or sub-strategies. U.S. 60/40 is 60% U.S. equities, 40% U.S. Treasuries. UH is unhedged. Results for 11 GICS equity sectors (unfilled teal markers) are based on data from February 1974 to June 2026, with correlations adjusted to match full sample by subtracting the difference in market correlations over the two samples. See Appendix for asset class proxies and construction of hypothetical portfolios. Hypothetical performance results have certain inherent limitations, some of which are disclosed in the Appendix.

One simple takeaway from Exhibit 4 is that not many investments “like” rising inflation. The chart highlights the impact of tilting from nominal to inflation-linked bonds (equivalent to adding exposure to breakevens) as well as the potential importance of energy equities, commodities and two types of trend following: price trend and economic trend. We’ll discuss these strategies in more detail later.²

Exhibit 5 compares long-term inflation sensitivity scores to performance during the 2022 bear market. Most asset classes performed as we would have expected (taking into account the different risk levels of, say, stocks and bonds). Gold, TIPS and breakevens disappointed as real yields soared but long-term inflation expectations remained anchored (though breakevens performed well in 2021). Commodities, trend following and long/short value delivered the strongest performance during the bear market.

Exhibit 5: What Have You Done for Me Lately? Long-Term Inflation Sensitivity vs. 2022 Performance



Source: AQR, Bloomberg, Survey of Professional Forecasters, U.S. Bureau of Labor Statistics. Inflation sensitivities are contemporaneous partial correlations to U.S. inflation metric, where the metric is an equal-risk combination of 12-month changes and 12-month surprises vs. survey forecasts. Based on rolling 12-month periods at quarterly frequency. U.S. 60/40 is 60% U.S. equities, 40% U.S. Treasuries. Return on vertical axis is total return from January 1 to September 30, 2022. See Appendix for asset class proxies and construction of hypothetical portfolios. Hypothetical performance results have certain inherent limitations, some of which are disclosed in the Appendix.

Which Kind of Inflation Matters?

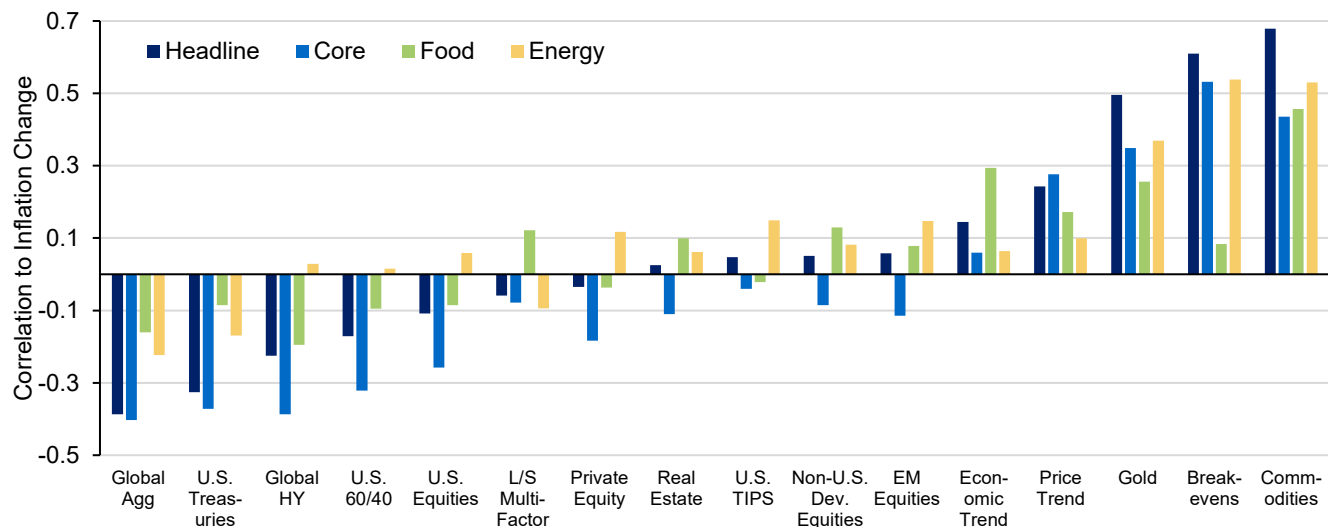
Our inflation news indicator is based on changes and surprises in headline CPI inflation, where we have long histories of forecasts. If we focus on year-on-year changes as a simpler news proxy, we can separate headline CPI into core, food and energy components. While energy price shocks can be dramatic, there is evidence that *rising core inflation* is the biggest risk to the economy and equity markets in the longer term.

Exhibit 6 confirms that equities, bonds, credit and illiquid assets are more sensitive to rising core inflation (mid-blue bars) than to food and energy inflation. Breakevens have delivered the most potent protection against core inflation shocks, but a diversified basket of commodities is also effective over this sample, as are the trend following strategies.

² In brief, price trend goes long or short each asset based on its own price trends (the classic trend following strategy), whereas as economic trend measures trends in macroeconomic variables and then goes long or short related assets. For example, an upward inflation trend would result in long commodity positions and short bond positions. Both strategies are based primarily on underreaction effects.

Exhibit 6: Sensitivities to Changes in Headline Inflation and Its Components

January 1, 1972 – June 30, 2026



Source: AQR, Bloomberg, U.S. Bureau of Labor Statistics. Sensitivities are two-factor partial contemporaneous correlations to U.S. inflation news metrics, where each metric is the 12-month change in the year-on-year rate, controlling for changes in GDP growth. Based on rolling 12-month periods at quarterly frequency. See Appendix for asset class proxies and construction of hypothetical portfolios. Hypothetical performance results have certain inherent limitations, some of which are disclosed in the Appendix.

Measuring Non-Linear Sensitivities

So far, we've reported sensitivities using linear correlation metrics. But what if investments exhibit different sensitivities to upside and downside inflation shocks? In **Exhibit 7** we use our metric to identify three regimes, when inflation is falling (blue), stable (gray), and rising (yellow). In each of these periods, Exhibit 7A reports the average return for selected asset classes, scaled to 10% volatility for ease of comparison. We see that while both stocks and bonds underperformed during inflationary periods, they had differing reactions when inflation news was negative. Bonds strongly outperformed in these environments, while stocks underperformed their average, displaying a non-linear "frown" pattern.³

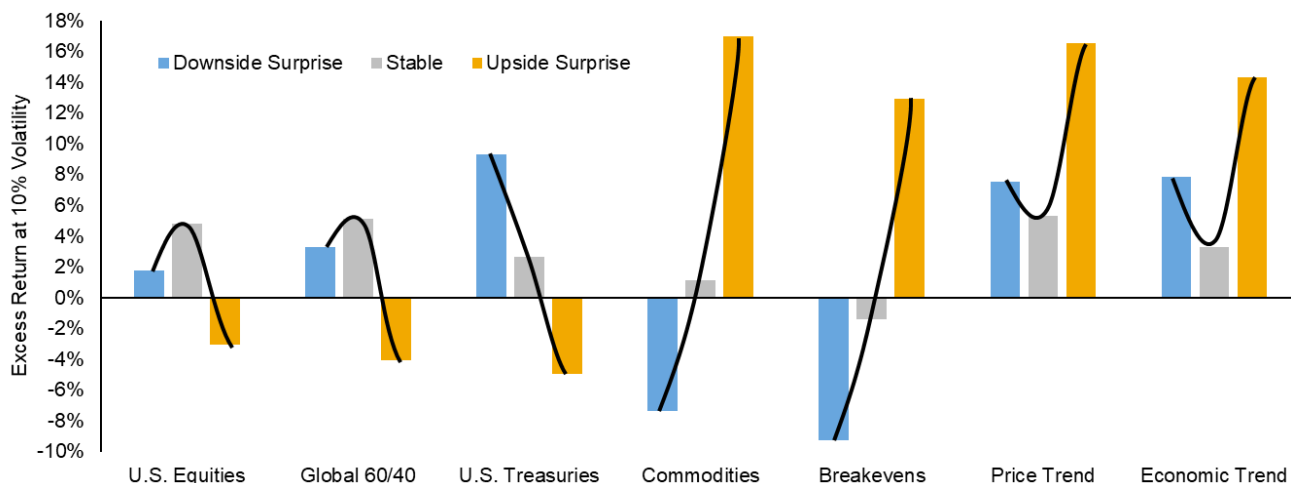
For commodities and breakevens, the relationship is close to linear, with strong outperformance in inflationary periods and strong negative performance in the opposite. This is a reminder that these assets may underperform in a disinflationary growth shock. An even better macro diversifier would be one that can outperform in both extremes, and this is where trend following strategies really shine. Since they can go long or short, both price and economic trend strategies have tended to outperform amid large shocks in either direction. This neatly diversifies equity-dominated portfolios, which prefer stable macro environments.

Exhibit 7B shows how recent rolling 12-month periods are categorized in this scheme. The period from 2024 to 2026 has seen largely stable inflation news, albeit drifting to the upside. This stable regime has previously been associated with the strongest equity market returns and the weakest trend performance. Investors should consider how to prepare for other regimes in the years ahead.

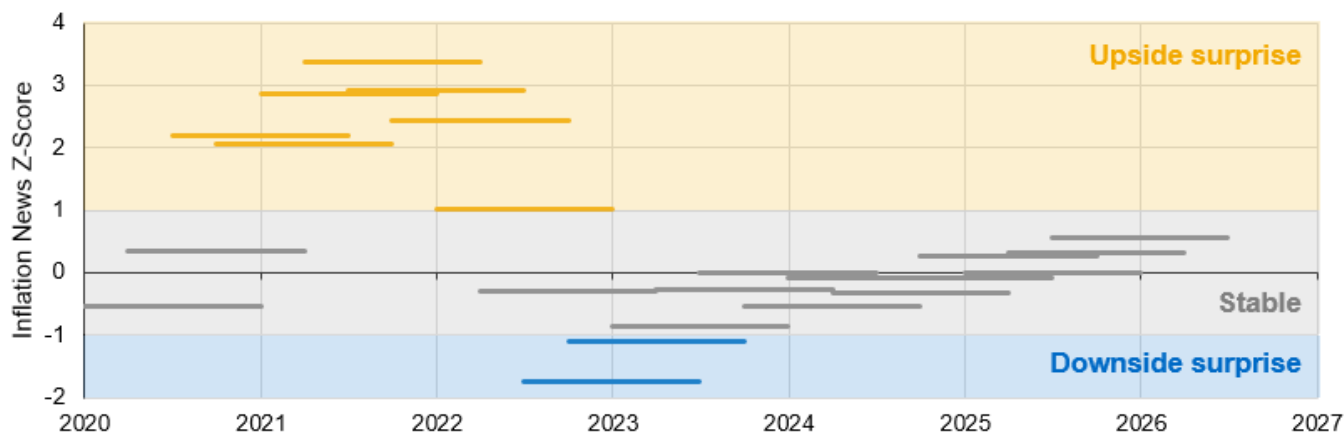
³ One possible cause of the 'frown' pattern for stocks is that adverse growth shocks may have coincided with (or caused) many of the disinflationary shocks in our sample. In the appendix we show upside and downside correlations controlling for growth, and find the non-linear pattern to be weaker but still visible. This remaining non-linearity may be the result of an imperfect proxy for growth, or caused by heightened risk aversion from investors in environments where the combination of economic shocks and policy responses has driven inflation away from central bank targets in either direction.

Exhibit 7: Frowns, Smirks and Smiles – Measuring Non-Linear Inflation Exposures

A. 3-Way Sort of Returns Relative to Inflation Environment Jan 1972 – Jun 2026



B. Recent Rolling 1-Year Inflation Regimes According to U.S. Inflation News Metric Jan 2020 – Jun 2026



Source: AQR, Bloomberg, Survey of Professional Forecasters, U.S. Bureau of Labor Statistics. Downside and upside surprise periods are defined as periods when inflation news metric has z-score <-1 and >1 respectively, with the rest of the sample categorized as stable. Asset class returns scaled to 10% volatility over full sample for ease of comparison. See Appendix for proxies and construction of hypothetical portfolios. Hypothetical performance results have certain inherent limitations, some of which are disclosed in the Appendix.

Inflation-Aware Asset Allocation

The Stock/Bond Correlation

We saw in Exhibit 5 that stocks and bonds have opposite-sign sensitivities to growth news, but same-sign sensitivities to inflation news. This means that when inflation uncertainty is elevated relative to growth uncertainty, the correlation between stocks and bonds tends to rise, episodes where both do poorly become more likely, and stock/bond investors find themselves with a diversification deficit (see Brixton et al. (2023)).

Exhibit 8 Chart A shows the negative stock/bond correlation during the first two decades of this century and the apparent regime change from 2022 onwards. Bonds can still provide important diversification in disinflationary growth shocks, but asset allocation analyses that anchor on the last 20 years may under-allocate to investments that can diversify in inflationary bear markets too. Chart A also highlights the rising correlation between credit and rates risk, which has historically been an important source of diversification for bond-heavy allocators like insurance companies. These investors are also facing a diversification deficit.

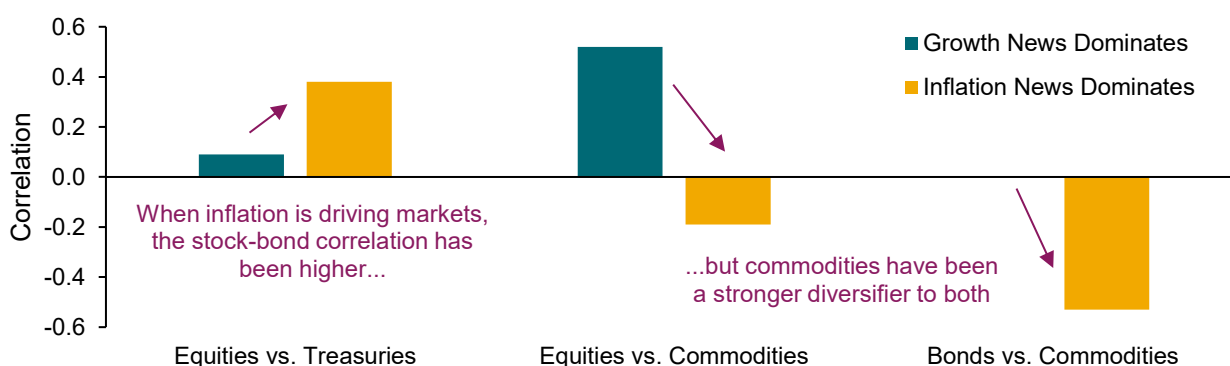
Chart B shows average correlations during two different macroeconomic regimes defined by comparing our growth and inflation news indicators. In periods when the magnitude of inflation news is larger (yellow bars), bonds are less diversifying to stocks, but commodities are strongly diversifying to both stocks and bonds. We'll apply this regime categorization to an asset allocation exercise shortly.

Exhibit 8: Asset Class Correlations Through Time and Across Macroeconomic Regimes

A. Rolling 5-Year U.S. Stock/Bond and Rates/Credit Correlations, Jan 1990 – Jun 2026



B. Correlations During Different Growth vs Inflation News Periods, Jan 1972 – Jun 2026



Sources: Bloomberg, Global Financial Data, Survey of Professional Forecasters, AQR. Upper chart uses rolling 60-month window of monthly returns; lower chart based on growth and inflation news indicators described previously, with rolling contemporaneous 12-month returns at overlapping quarterly frequency. Sample is divided into periods when magnitude of inflation news was bigger/smaller than growth surprise (ignoring sign of news), as a proxy for relative uncertainty. See Appendix for asset class proxies.

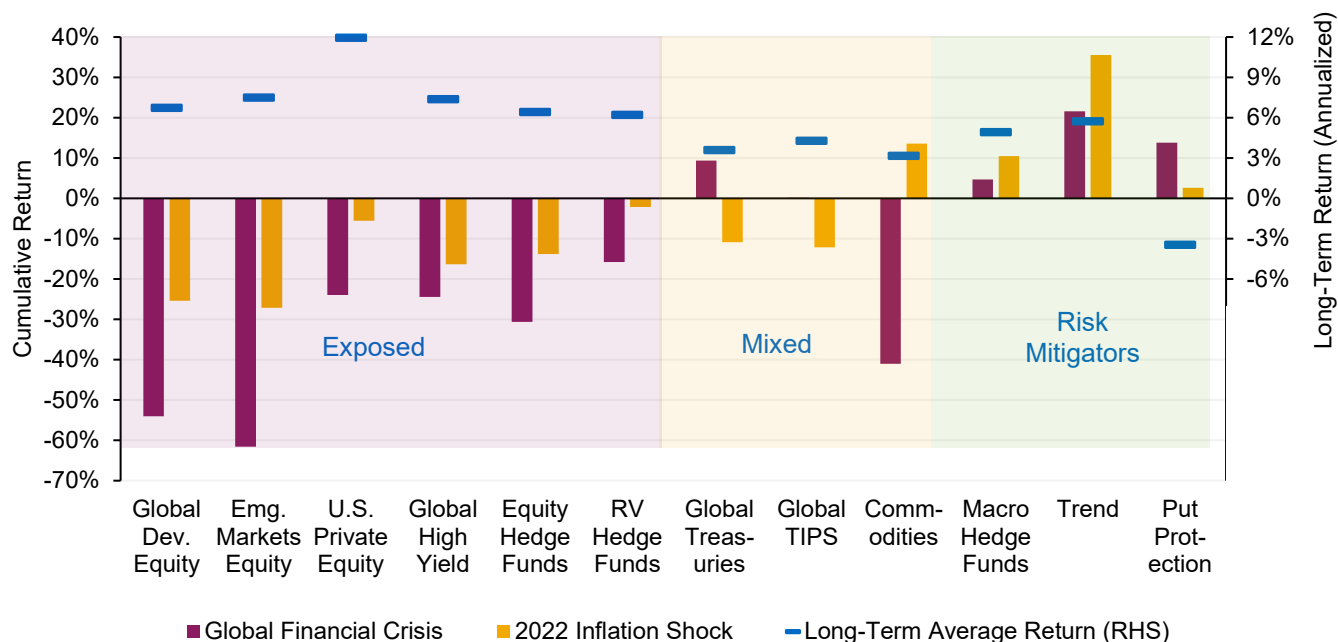
Inflationary and Disinflationary Tail Risks

Asset allocation frameworks with a downside risk element that focuses solely on the Global Financial Crisis (GFC, often the worst historical drawdown in the chosen data set) are likely to underemphasize inflation risk because this was a disinflationary growth shock. For earlier inflationary bear markets one had to go back to the 1970s, until 2022 provided a much more recent example. By including analysis of both the GFC and the 2022 drawdown when constraining portfolio risk, investors can be more confident of understanding their downside risk in a wider range of adverse macroeconomic outcomes.

Exhibit 9 highlights the performance of 12 asset classes and strategy types during these two episodes. The opposite performance of nominal bonds and commodities is particularly striking, and is no surprise given their opposite inflation sensitivities in Exhibit 4. The most reliable risk mitigators, in the green zone, were

macro hedge funds, trend following strategies and equity index put options. However, while option-based protection has incurred a substantial long-term cost, trend following has delivered positive returns (blue lines). Trend following is likely to be a useful allocation in analyses that incorporate both inflationary and disinflationary tail risk episodes.

Exhibit 9: Performance in Disinflationary and Inflationary Bear Markets January 2000 – June 2026



Sources: Bloomberg and AQR. Based on monthly total returns. Global Financial Crisis period is November 1, 2007 to March 31, 2009. 2022 Inflation Shock period is January 1, 2022 to September 30, 2022. Long-term return is geometric mean calculated over period January 1, 2000 to June 30, 2026. See Appendix for asset class proxies. Past performance does not predict future returns.

Robust Asset Allocation Case Study

Let's consider an asset allocation example that accounts for both growth-dominant and inflation-dominant regimes, and both disinflationary and inflationary tail events. We start from a global 60/40 stock/bond portfolio. We then introduce three potential inflation diversifiers – inflation linked bonds, commodities and trend following – and conduct two constrained optimizations based on different correlation assumptions and different tail risks, before combining the results to produce our “robust” optimal portfolio.

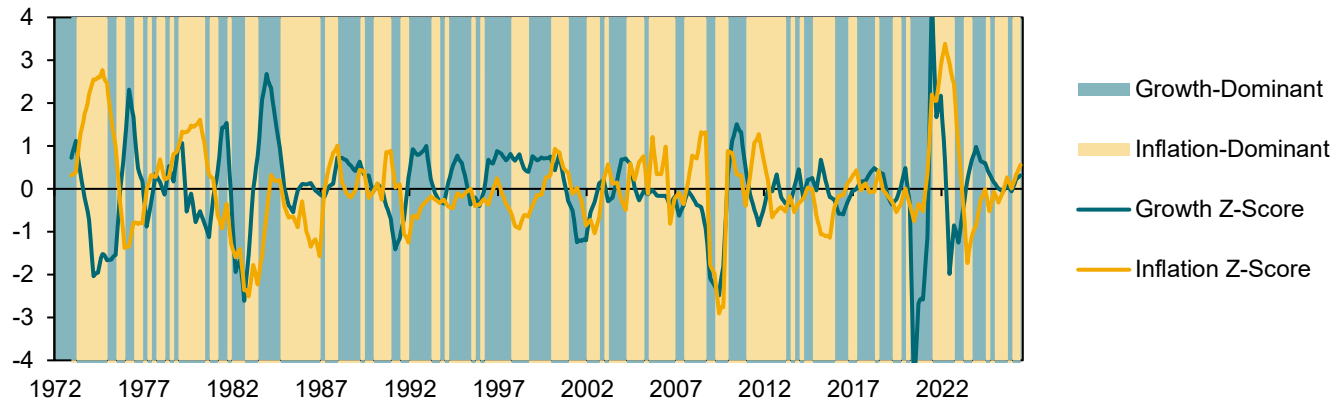
Both optimizations seek to maximize expected portfolio compound total return based on a set of forward-looking assumptions and subject to various constraints. Expected portfolio volatility must not exceed that of the starting portfolio, and the rolling 12-month historical underperformance must not exceed 5%.⁴ The loss during two historical tail events – one disinflationary and one inflationary – must not exceed that of the starting portfolio. A modest 15% leverage is permitted.⁵

⁴ The latter constraint effectively limits how much the optimal allocation can differ from the starting portfolio, and is intended to represent active risk, peer risk or conventionality constraints which may be explicit or implicit. Investors could also constrain expected tracking error (reported in the results) or use aversion terms rather than hard constraints.

⁵ This does not require the allocator to take on-balance-sheet leverage. The same effects can be achieved through a variety of capital efficient investment solutions (higher-duration bonds, risk parity, active extension equity, portable alpha, and so on). See AQR Alternative Thinking 2025 Issue 3: Exploring Capital Efficiency for further discussion.

The growth-dominant optimization uses correlations experienced in growth-dominant environments, as defined by our growth and inflation news metrics, shown in **Exhibit 10**. The inflation-dominant optimization uses correlations experienced in inflation-dominant environments. Risk and return assumptions are the same in both optimizations and are given in the appendix.

Exhibit 10: Dividing Our Sample to Generate Two Sets of Assumptions Jan 1972 – Jun 2026



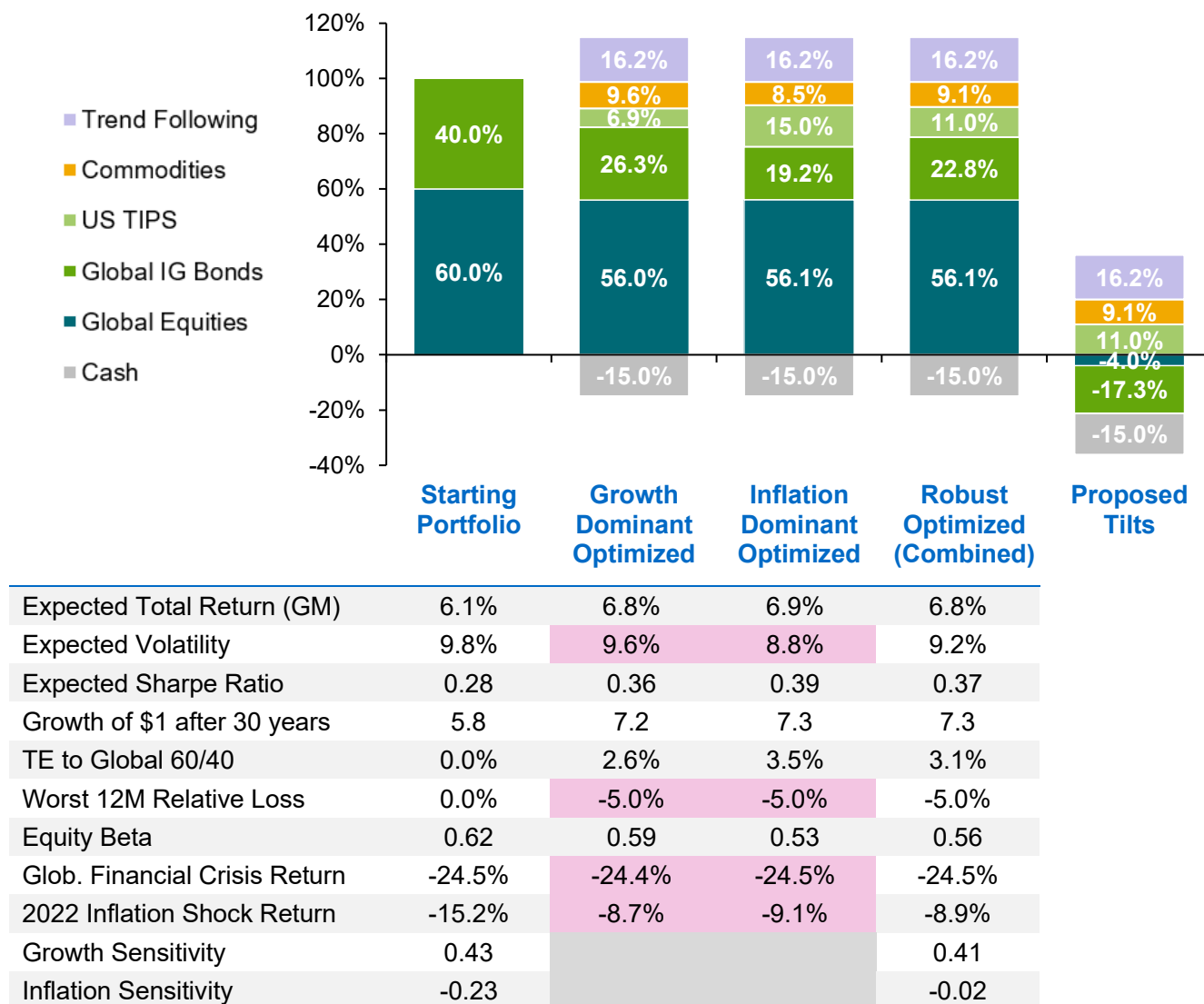
Sources: Bloomberg, Global Financial Data, Survey of Professional Forecasters, AQR. Growth-dominant regime is when absolute z-score of growth news indicator is greater than absolute z-score of inflation news indicator.

Exhibit 11 shows the results of the two optimizations, the combined portfolio and the implied changes to the starting portfolio. The two optimal allocations are similar because the tail risk, relative loss and leverage constraints are binding on both. Both seek to replace some nominal bonds with more capital-efficient diversifiers of equity risk; both use the permitted leverage to boost diversification; and both are limited in their commodity allocation by the need to meet the disinflationary tail constraint (GFC performance). There are some differences because of the different diversification assumptions. For example, the inflation dominant optimal portfolio has a larger re-allocation from nominal to inflation-linked bonds (equivalent to adding breakeven exposure), which are more diversifying in this regime. Trend following is favored under both sets of assumptions and helps reduce both tail event losses, but it does incur tracking error and periods of underperformance.

The combined portfolio allocates to inflation-linked bonds, commodities, and trend following, funded from a blend of public equity, nominal bonds and leverage. One way to implement the leverage would be to allocate to trend following via a portable alpha strategy that also provides equity market exposure.

According to our assumptions, this combined optimal portfolio has an expected return around 70bps higher than the starting portfolio with a similar level of risk and similar growth sensitivity, but with substantially reduced inflation sensitivity and improved performance in the inflationary tail event. Importantly, the optimization considers both growth-dominant and inflation-dominant regimes, and avoids the tendency of some allocation analyses to overemphasize disinflationary assumptions and tail risks.

There are many ways to approach this analysis – one could, for example, allocate to an explicit inflation hedge portfolio – but the resulting re-allocation proposals are likely to remain broadly the same.

Exhibit 11: Optimal Portfolios With Different Assumptions

Sources: Bloomberg, Global Financial Data, Survey of Professional Forecasters, AQR. Optimized portfolios maximize expected total return subject to constraints on volatility, relative 12-month performance, tail event returns and leverage. Pink shading indicates constrained parameters. GM is geometric mean; TE is tracking error. Portfolio return and risk expectations are based on assumptions given in the appendix. Global Financial Crisis period is November 1, 2007 to March 31, 2009. 2022 Inflation Shock period is January 1, 2022 to September 30, 2022. Growth and inflation sensitivities are long-term partial correlations to the metrics defined earlier. Risk statistics for starting portfolio and robust optimized portfolio are averaged across the two regimes.

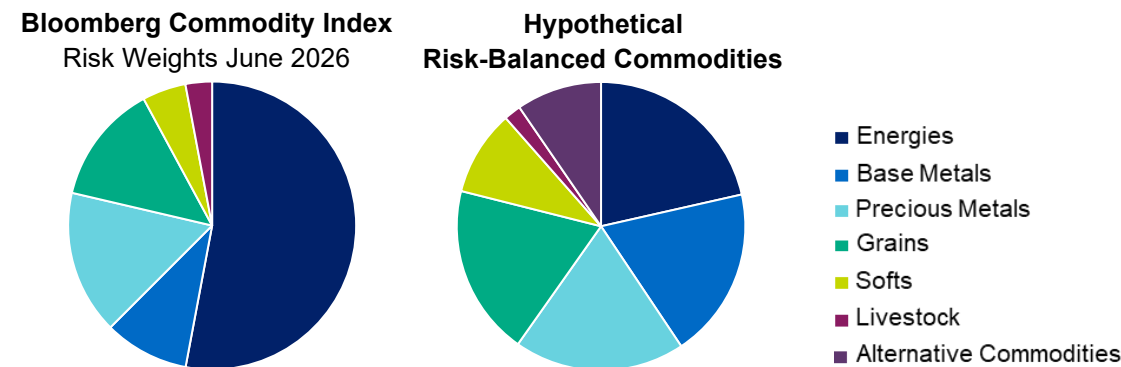
Implementation: Optimizing Resilience/Return Trade-Offs

We've seen that allocations to commodities and trend following may be able to increase a portfolio's inflation resilience without reducing long-term performance, but they will introduce tracking error and require organizational commitment. It's the investor's job to select an implementation that optimizes these trade-offs. In this section we explore, for each strategy, implementation considerations that may help to deliver the best combination of macroeconomic resilience and long-term return.

Commodities

Three key considerations for a commodities allocation are (1) strategic sector mix, (2) risk management, and (3) active management. The popular indices are dominated by exposure to the energy sector, as shown in **Exhibit 12**.⁶ This is useful for generating returns during energy shocks, but not so good for maximizing long-term risk-adjusted returns or diversifying active opportunities. In the appendix we show inflation sensitivities for the different sectors and the perhaps surprising result that a sector-balanced portfolio has a similar inflation sensitivity as an energy-dominated mix. Holding a balanced sector allocation will create some tracking error versus the popular indices, but we expect it will be well compensated over the long term, without loss of inflation protection.

Exhibit 12: Traditional Commodity Indices are Concentrated in Energies



Source: Bloomberg, AQR. Charts are for illustrative purposes only. Risk allocations based on 2026 sector weights for Bloomberg Commodity Index, long-term strategic weights of a hypothetical risk-balanced commodities portfolio described in the Appendix, and AQR volatility and correlation estimates. The strategic risk weights shown for a hypothetical risk-balanced commodities portfolio are consistent with the hypothetical portfolio described in the Appendix and are for illustrative purposes only. Hypothetical data has inherent limitations, some of which are disclosed in the Appendix. Diversification does not eliminate the risk of experiencing investment losses.

The risk profile of a commodities allocation can be improved by dynamic risk targeting – adjusting notional exposures through time – and long-term returns can be enhanced through active management. **Exhibit 13** summarizes active investment themes in commodity markets, which can be applied in absolute return strategies, long-biased and long-only portfolios. Investors should seek a commodities strategy that is well diversified across sectors but also takes advantage of these active opportunities.

Exhibit 13: Active Investment Themes in Commodity Markets

Carry	Slope of futures curves may reflect risk premia (hedging pressure) as well as inventories
Momentum	Momentum seeks to take advantage of trends in commodity markets
Seasonal Risk Premia	Commodities exhibit time varying risk premia, which may be systematically harvested
Macroeconomic Drivers	Macroeconomic variables of commodity-consuming countries drive commodity prices
Supply/Demand	Variety of fundamental data which measure or predict supply / demand dynamics
Value	Reversal and mean reversion factors take advantage of substitution and overreaction

Source: AQR.

⁶ In Exhibit 10, we show the Bloomberg Commodity Index. The other most popular commodity index, the S&P GSCI, is production-weighted and tends to be even more concentrated in energy sector risk.

Finally, **Exhibit 14** summarizes the performance of a hypothetical risk-balanced commodities strategy with an active component. The return attribution shows that the strategic risk balance and the tactical signals both contributed to the strategy's substantial outperformance of its benchmark.

Exhibit 14: Hypothetical Active Risk-Balanced Commodities vs. Passive Index

May 1, 1990 – June 30, 2026

	Hypothetical Risk-Balanced Commodities	Bloomberg Commodity Index	Excess-of- Benchmark Return	Excess Return Attribution	
				Strategic Allocation	Active Tilts
Total Return (GM)	11.9%	3.6%			
Excess Ret (AM)	10.0%	1.8%	8.2%	1.7%	6.4%
Volatility	17.4%	14.8%	8.1%		
Sharpe Ratio / IR	0.57	0.12	1.01		

Source: AQR. Performance from May 1, 1990 through June 30, 2026 of a hypothetical risk-balanced commodities portfolio (described in the Appendix) and the Bloomberg Commodity Index. Performance is in USD, net of transaction costs and gross of fees. AM and GM are arithmetic and geometric mean. IR is information ratio. Past performance is not a reliable indicator of future performance. Please read important disclosures in the Appendix. The risk-free rate used to calculate the Sharpe ratios is the ICE BofAML 3-month U.S. Treasury Bill. Hypothetical data has inherent limitations, some of which are disclosed in the Appendix.

Trend Following

Trend following strategies are designed to capitalize on large and prolonged market moves – these can be in either direction and in any liquid asset class, and they often occur when there is a macroeconomic shock.⁷ But not all trend strategies are alike in their ability to improve portfolio resilience, or in their ability to generate positive returns over the long term. Two key considerations are (1) the asset universe and asset class allocation, and (2) the signals employed.

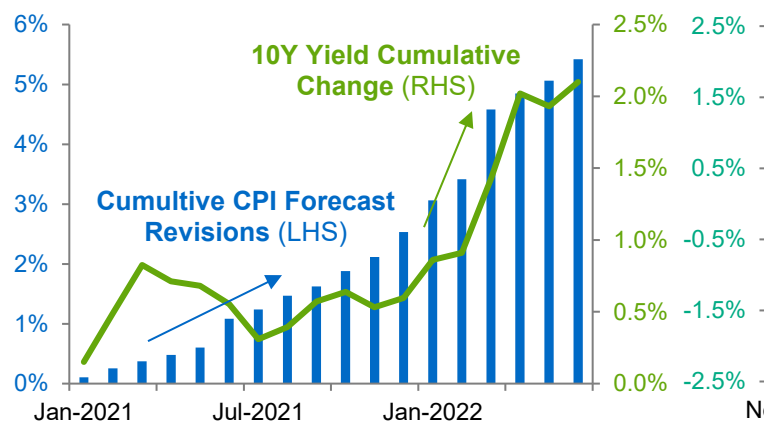
Many trend strategies trade a similar set of assets across major equity, bond, currency and commodity markets, but not all have a balanced risk allocation across these asset classes. A few offer a much broader universe that may include smaller commodity markets, digital assets, emerging market rates and even long/short equity factors. These may help the strategy to deliver a more consistent return profile, while maintaining its tail risk mitigation capabilities.

An increasing number of trend strategies exploit a wider range of signals beyond the typical signals identifying price trends. Some of these “complementary signals” may help to generate returns in calm markets but at the cost of diluting the macroeconomic convexity of the strategy (carry signals and option-writing strategies are two examples). Economic trend signals are different. They seek to identify trends in macroeconomic fundamentals including inflation and monetary policy, and can actually enhance the convexity of the broader trend strategy, as well as providing a new source of alpha. **Exhibit 15** gives two examples of how economic trend signals contributed to strong performance during the 2022 inflation shock: identifying the rising inflation trend before bond markets reacted (Chart A) and identifying the consequent policy tightening and its likely negative impact on equity markets (Chart B).

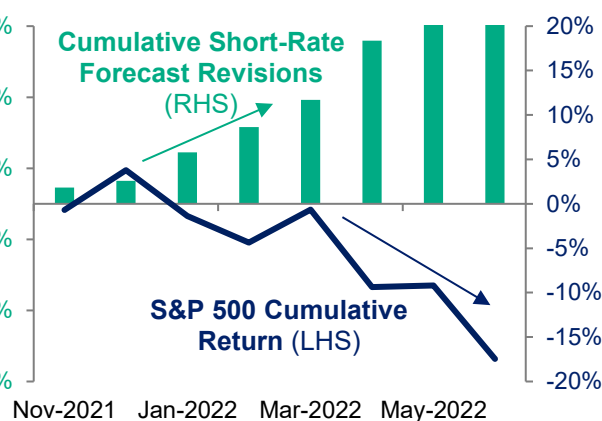
⁷ Some shocks can cause market reversals that are challenging for trend followers in the shorter term. But the largest shocks and deepest bear markets have tended to occur over prolonged periods and have been favorable environments for trend strategies. See for example McQuinn et al. (2021).

Exhibit 15: How Economic Trends Helped to Predict Market Trends During 2022

A. U.S. CPI Inflation Forecast Revisions vs. Treasury Yield Changes, Jan 2021 - Jun 2022



B. U.S. Short-Rate Forecast Revisions vs. S&P 500 Returns, Nov 2021 - Jun 2022

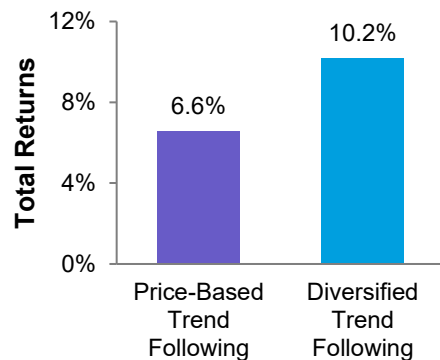


Source: AQR, Bloomberg. Past performance is not a reliable indicator of future performance.

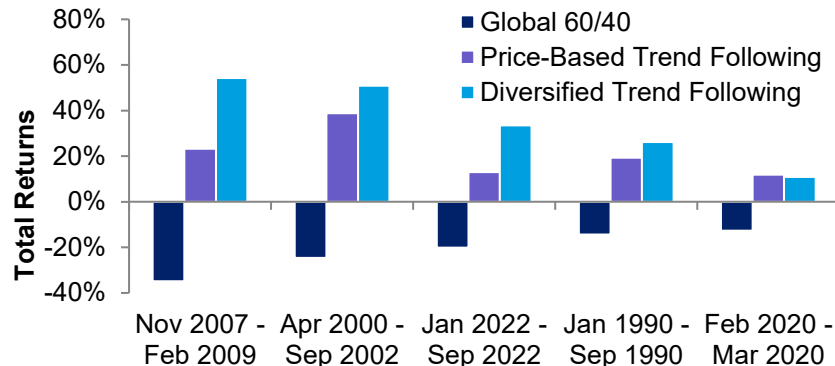
Adding a wider universe of alternative assets and incorporating economic trend signals are two important ways to improve the long-term performance of trend following strategies, while maintaining their ability to mitigate major portfolio risks including inflation shocks. **Exhibit 16** summarizes backtest evidence for both higher risk-adjusted returns and enhanced risk mitigation potential for a Diversified Trend strategy that includes these innovations.

Exhibit 16: Hypothetical Benefits of Adding Alternative Markets and Economic Trend

A. Broader Sources of Alpha January 1990 – June 2026



B. Enhanced Risk Mitigation: 5 Worst Drawdowns for Global 60/40 January 1990 – June 2026



Source: AQR, Bloomberg. Hypothetical Diversified Trend-Following Strategy performance is a backtest that is 40% Price-Based Trend Following, 40% Economic Trend Following and 20% Alternative Trend Following. Backtest targets 10% volatility and is net of estimated transaction costs and net of 1.25 & 20 fees. See appendix for a description of the investment universe and the allocation methodology used to construct the hypothetical strategies. Global 60/40 is 60% MSCI World USD and 40% Bloomberg Global Aggregate. Charts for illustrative purposes and not based on a portfolio AQR manages. Hypothetical data has inherent limitations, some of which are in the Disclosures. Diversification does not eliminate the risk of experiencing investment losses.

Concluding Thoughts

Investors should be cautious about aggressive tactical bets on the direction of inflation, but inflation-related tails risks should not be ignored. Our empirical analysis consistently suggests that strategic allocations to inflation-linked bonds, commodities and trend following may help to improve portfolio resilience at a time of

uncertain inflation outlook. Investors should pay heed, and seek out the best implementations of these strategies to do the job. Trend following in particular is a versatile risk mitigator, having outperformed in both inflationary and disinflationary bear markets.

Our analysis has focused on inflation surprises over a 12-month horizon, but the dangers of longer-term inflation expectations becoming de-anchored seem real in late 2026, with challenges to the Fed's independence coinciding with loose fiscal policies and unpredictable trade policies. Rising long-term inflation expectations could have a "triple-whammy" impact on Treasury yields: the direct 1:1 impact on nominal yields, a higher inflation risk premium, and a higher term premium as the higher stock-bond correlation hurts Treasuries' role as safe-haven assets. At some point these outcomes would be harmful to risky assets too: what happens in Treasuries does not stay in Treasuries. Looking back to the last episode of de-anchored inflation expectations in the 1970s, we may not be surprised to see that two of the best-performing strategies were, once again, commodities and trend following.

References and Further Reading

AQR *Alternative Thinking* 2020 Q4: Fire and Ice: Confronting the Twin Perils of Inflation and Deflation, AQR white paper.

AQR *Alternative Thinking* 2025 Issue 3: Exploring Capital Efficiency, AQR white paper.

Babu, A., A. Levine, Y. Ooi, L. Pedersen, and E. Stamelos, 2020, "Trends Everywhere," *Journal of Investment Management*, 18(1).

Brixton, A., T. Maloney, and A. Thapar, 2021, "When Stock-Bond Diversification Fails: Managing Inflation Risk in Investor Portfolios," AQR white paper.

Brixton, A., J. Brooks, P. Hecht, A. Ilmanen, T. Maloney, and N. McQuinn, 2023, "A Changing Stock–Bond Correlation: Drivers and Implications," *Journal of Portfolio Management*, 49(4).

Brooks, J., Y. Ooi, N. Feilbogen, and A. Akant, 2023, "Economic Trend," AQR white paper.

Ilmanen, A., T. Maloney, and A. Ross, 2014, "Exploring Macroeconomic Sensitivities: How Investments Respond to Different Economic Environments," *Journal of Portfolio Management*, 40(3).

Ilmanen, A., 2022, "Investing Amid Low Expected Returns," Wiley.

Ilmanen, A. and T. Maloney, 2025, Understanding Return Expectations Part 8: Understanding Treasury Yields with Survey Data, AQR white paper.

Levine, A., Y. Ooi, M. Richardson and C. Sasseville, 2018, "Commodities for the Long Run," *Financial Analysts Journal*, 74(2).

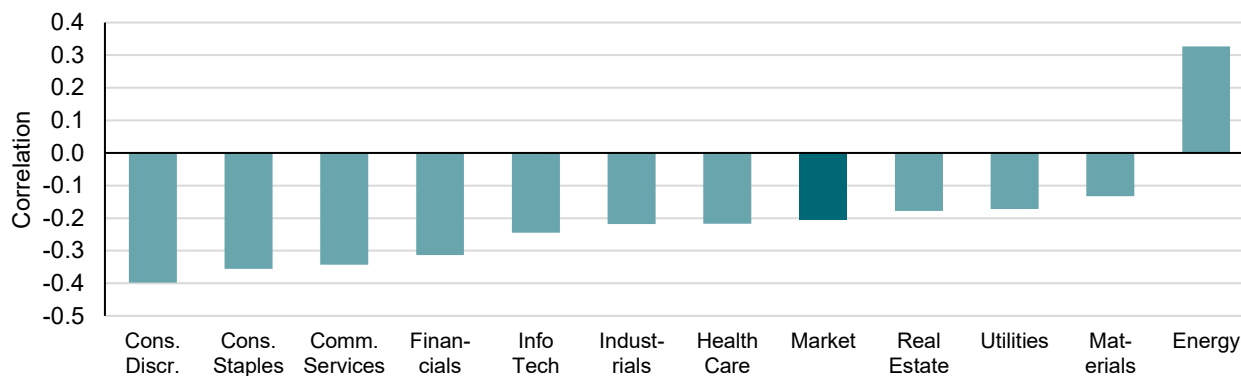
McQuinn, N., A. Thapar and D. Villalon, 2021, "Portfolio Protection? It's a Long (Term) Story," *Journal of Portfolio Management*, 47(3).

Ooi, Y., T. Maloney and A. Brixton, 2022, "Building a Better Commodities Portfolio", AQR white paper.

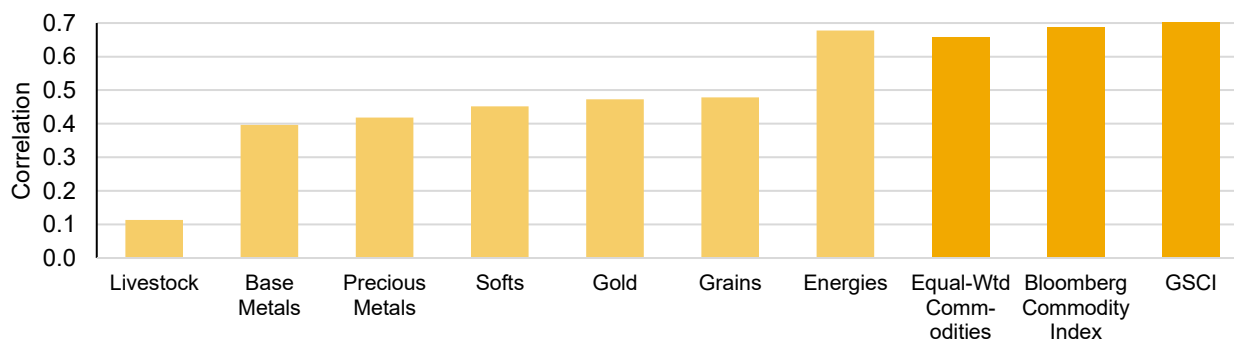
Appendix

Exhibit A1: Inflation Sensitivities for Equity and Commodity Sectors

A. U.S. Equity Sectors, February 1974 – June 2026



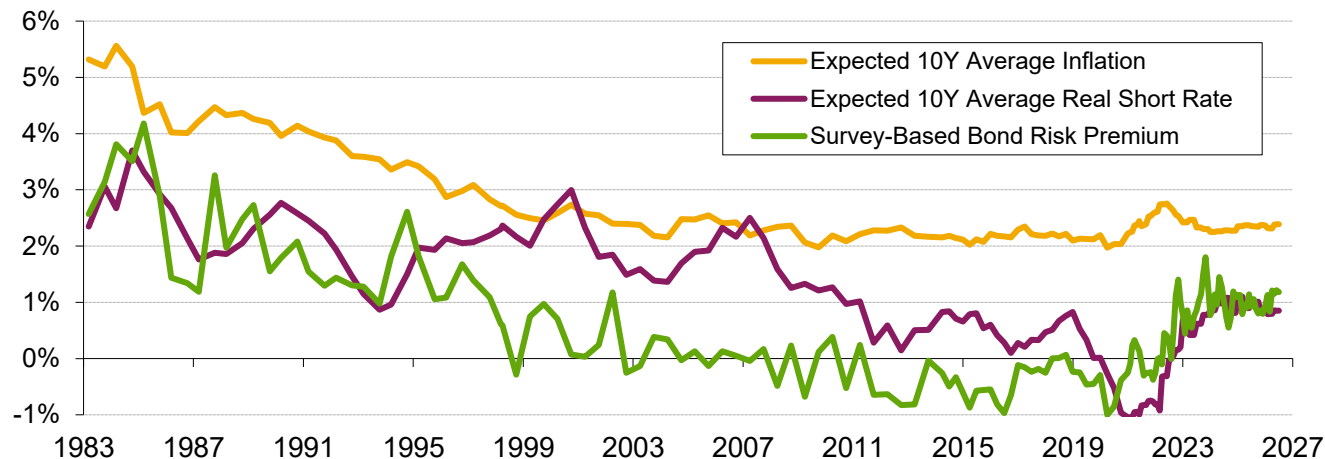
B. Commodity Sectors and Portfolios, January 1972 – June 2026



Source: AQR, Bloomberg, Survey of Professional Forecasters, U.S. Bureau of Labor Statistics. Sensitivities are simple correlations to inflation metric described in main text. Base metals data start only in 1977, energies in 1983. See below for details of asset class proxies.

Exhibit A2 uses survey data to decompose the 10-year U.S. Treasury yield into three parts: expected average inflation, the expected average real short rate, and the bond risk premium (BRP, see Ilmanen and Maloney (2025) for more details). It was the real rate and the BRP that increased sharply after 2022.

Exhibit A2: The Three Components of the 10-Year Treasury Yield January 1983 – June 2026



Sources: AQR, FRED database, Blue Chip Economic Indicators, Consensus Economics. Besides the actual 10-year Treasury yield, the chart includes consensus forecasts of next-decade average 3-month Treasury bill rate and the consumer price inflation rate.

Asset Class Proxies for Inflation Sensitivity Analysis (Exhibits 4-8)

Asset Class	Proxy 1	Start Date	Proxy 2	Start Date	Proxy 3	Start Date
U.S. Equity	MSCI USA Net TR Index	1/1/1972				
Non-U.S. Dev. Equity	MSCI World ex US Net TR Unhedged Index	1/1/1972				
Emerging Equity	MSCI Emerging Markets Net TR Unhedged Index	1/1/1988	MSCI EAFE Net TR Unhedged Index	1/1/1972		
U.S. Treasuries	U.S. 10-year Treasury	1/1/1972				
Global IG Bonds	Bloomberg Global Aggregate Bond Index Hedged	2/1/1990	AQR Global Bond Series (volatility adjusted)	1/1/1972		
High Yield Bonds	Bloomberg US High Yield Index	8/1/1983	Bloomberg US Corporate Bond Index	2/1/1973	AQR US Bond Series	1/1/1972
Credit Excess	Bloomberg US Corporate Bond Credit Excess Index	8/1/1988	Credit return from Asvanunt and Richardson (2016)	1/1/1972		
Inflation-Linked Bonds	U.S. 10-Year TIPS	3/1/1997	Synthetic returns based on nominal Treasury yields and survey-based expected inflation	1/1/1972		
Private Equity	50% Russell 2000 x 1.2 (levered listed small cap index), 50% Cambridge U.S. Buyout (unlisted)	4/1/1986	70% Russell 2000 x 1.2 (levered listed small cap index), 30% Cash	1/1/1972		
Real Estate	50% FTSE NAREIT All REITs Index (listed), 50% NCREIF Index (unlisted)	12/1/1977	60% FTSE NAREIT All REITs Index (listed), 40% Cash	1/1/1972		
Commodities	Bloomberg Commodities Index	1/1/1972				
Gold	Bloomberg Gold TR Index	1/1/1972				
B/E Inflation	Long 10-year U.S. TIPS as defined above, short 10-year U.S. Treasury	1/1/1972				
Hypothetical Long/Short Equity Factors	Value is U.S. HML Devil factor based on B/P (Asness and Frazzini, 2013). Momentum is U.S. 'Up Minus Down' factor based on 12M return excluding most recent month. Low Beta is U.S. 'Betting Against Beta' factor (Frazzini and Pedersen, 2014). Quality is U.S. 'Quality Minus Junk' factor (Asness, Frazzini and Pedersen, 2014). All factors are cap-weighted long the 1/3 best stocks and short the 1/3 worst stocks following the methodology of Fama and French (1993), rebalanced annually every January. All factors except Low Beta are dollar long and short. For Low Beta the long side is levered to make the portfolio ex-ante beta-neutral as described in Frazzini and Pedersen (2014). Returns are discounted by 33% with breakpoint in 2010, and net of 1% management fee. Multi-factor Portfolio is equal-weighted combination of Value, Momentum and Defensive, where Defensive is 50% Low Beta and 50% Quality. Multi-factor Portfolio returns are also discounted and are net of 2% management fee.					
Hypothetical Price Trend	The Hypothetical Price-Based Trend-Following Strategy is equal-weighted combination of 1-month, 3-month, and 12-month trend-following strategies for 67 markets across 4 major asset classes: 29 commodities, 11 equity indices, 15 bond markets, and 12 currency pairs. We use futures returns when they are available. Prior to the availability of futures data, we rely on cash index returns financed at local short rates for each country. The strategy targets a long-term volatility target of 10%. Returns are net of estimated transaction costs and 1.25% & 20% fees.					
Hypothetical Economic Trend	The Hypothetical Economic Trend Strategy is based on trend following which for each theme (Growth, Inflation, International Trade, Monetary Policy, Risk Aversion) and within each asset class, takes a long position in assets in which economic trends are improving and a short position in assets in which economic trends are deteriorating. Each individual position is sized to target the same amount of volatility, both to provide diversification and to limit the portfolio risk from any individual market. The theme portfolio across all assets is scaled to target 10% forecasted annual volatility. Returns are net of estimated transaction costs and 1.25% & 20% fees.					
Cash	Carry on 3-month U.S. T-Bills	1/1/1972				

Asset Class Proxies for Tail Risk Analysis (Exhibit 9)

Asset Class	Proxy
Global Developed Equities	MSCI World USD Index
Emerging Market Equities	MSCI Emerging Markets USD Index
US Private Equity	Cambridge Associates Private Equity Index
Global HY Bonds	Bloomberg Global High Yield Index (hedged)
Equity Hedge Funds	HFRI Equity Hedge Index
RV Hedge Funds	HFRI Relative Value Index
Global Treasuries	Bloomberg Global Treasury Index (hedged)
Global TIPS	Bloomberg Global TIPS Index (hedged)
Commodities	Bloomberg Commodity Index
Macro Hedge Funds	HFRI Macro Index
Trend Following	SG Trend Index
Put Protection	PPUT Index minus S&P 500 Index

Assumptions for Asset Allocation Analysis (Exhibit 11)

Expected returns are yield-based estimates as of June 30, 2026, with methodologies as described in AQR *Alternative Thinking* 2026 Issue 1: Capital Market Assumptions for Major Asset Classes. Volatilities are based on monthly proxy data. Sharpe ratios are calculated from expected returns and volatilities. Correlations are based on longer-term proxies since 1972 as listed on previous page, with regimes defined in the main text. GM is geometric mean.

	Total GM Return	Volatility	Sharpe Ratio	Proxy for Volatility
Global Equities	6.5%	15.6%	0.25	MSCI World USD
Global IG Bonds	4.7%	3.4%	0.27	Bloomberg Global Aggregate
Global TIPS	5.0%	5.6%	0.24	Bloomberg US TIPS Index
Commodities	6.9%	15.6%	0.27	Bloomberg Commodities Index
Trend Following	6.1%	12.3%	0.25	SG Trend Index
Cash	3.8%			

Correlations		Global Equities	Global IG Bonds	Global TIPS	Commodities	Trend Following
Growth Dominant	Global Equities	1.00				
	Global IG Bonds	0.09	1.00			
	Global TIPS	0.15	0.77	1.00		
	Commodities	0.52	0.00	0.22	1.00	
	Trend Following	-0.26	0.02	-0.09	0.09	1.00
Inflation Dominant	Global Equities	1.00				
	Global IG Bonds	0.38	1.00			
	Global TIPS	0.09	0.60	1.00		
	Commodities	-0.19	-0.53	0.02	1.00	
	Trend Following	-0.26	-0.18	0.10	0.47	1.00

Hypothetical Risk-Balanced Commodities and Diversified Trend Strategies (Exhibits 14 & 16)

The **Hypothetical Risk-Balanced Commodities Strategy** rebalances monthly and is risk-balanced across the following 26 commodities in 6 sectors: Agriculturals (Corn, Soybeans, Wheat, Soybean Meal, Soybean Oil), Softs (Coffee, Cocoa, Cotton, Sugar), Energies (WTI Crude, Brent Crude, Gas Oil, Gasoline, Heating Oil, Natural Gas), Livestock (Live Cattle, Feeder Cattle, Lean Hogs), Precious Metals (Gold, Silver), Base Metals (Aluminum, Platinum, Copper, Nickel, Zinc, Lead). It gives equal notional weight to commodities within each sector. The portfolio targets a constant volatility, subject to a risk reduction process. The portfolio volatility is estimated using a combination of individual volatilities and pair-wise correlations. The individual volatilities are calculated using historical daily returns, weighted exponentially with a 60-day center of mass. Correlation estimates are calculated using 5-day overlapping returns (to account for different market close times), exponentially weighted with a 150-day center of mass.

The **Hypothetical Diversified Trend-Following Strategy** has a 40% allocation to the Hypothetical Price-Based Trend-Following Strategy, a 40% allocation to the Hypothetical Economic Trend-Following Strategy and a 20% allocation to the Hypothetical Alternative Trend-Following Strategy. This backtest is then scaled to target 10% volatility. Please see below for descriptions of each of the components of this backtest. Returns are net of estimated transaction costs and 1.25% & 20% fees.

The **Hypothetical Price-Based Trend-Following Strategy** goes long markets that have been rising and short markets that have been falling, betting that those trends over the examined look-back periods will continue. The strategy was constructed with an equal-weighted combination of 1-month, 3-month, and 12-month trend-following strategies for 67 markets across 4 major asset classes: 29 commodities, 11 equity indices, 15 bond markets, and 12 currency pairs. We construct the strategies using the largest number of assets for which return data exist at each point in time. We use futures returns when they are available. Prior to the availability of futures data, we rely on cash index returns financed at local short rates for each country. Please see table below for additional details. The strategy targets a long-term volatility target of 10% but does not limit volatility during periods where realized volatility may be higher or lower than this number.

Hypothetical performance is net of estimated transaction costs and net of 1.25% annual management fee and 20% performance fee. The transactions costs used in the strategy are based on proprietary estimates of average transaction costs for each of the four asset classes, including market impact and commissions, and are assumed to be twice as high from 1993 to 2002 and six times as high before 1993. The transaction costs used are shown to the right. This model is not based on an actual portfolio AQR manages.

Asset Class	Time Period	One-Way Transaction Costs (as % of notional traded)
Equities	1880 – 1992	0.34%
	1993 – 2002	0.11%
	2003 – Present	0.06%
Fixed Income	1880 – 1992	0.06%
	1993 – 2002	0.02%
	2003 – Present	0.01%
Currencies	1880 – 1992	0.18%
	1993 – 2002	0.06%
	2003 – Present	0.03%
Commodities	1880 – 1992	0.58%
	1993 – 2002	0.19%
	2003 – Present	0.10%

The **Hypothetical Economic Trend-Following Strategy** is based on trend following which for each theme (Growth, Inflation, International Trade, Monetary Policy, Risk Aversion) and within each asset class, takes a long position in assets in which economic trends are improving and a short position in assets in which economic trends are deteriorating. Each individual position is sized to target the same amount of volatility, both to provide diversification and to limit the portfolio risk from any individual market. The theme portfolio across all assets is scaled to target 10% forecasted annual volatility. Not all markets and assets have returns going back to 1970; details outlined later in this description. Themes are as follows:

Growth: Growth trends are captured using one-year changes in forecasts of real GDP growth. From 1990 onward forecast data is from Consensus Economics. Prior to 1990, we use one-year changes in realized year-on-year real GDP growth, lagged one quarter (this definition is equivalent to changes in forecasts assuming that real GDP growth follows a random walk). The series is from the OECD. Increasing growth is assumed to be bullish for equities (cash-flow impact), commodities (increasing demand), and currencies (Balassa-Samuelson hypothesis), and bearish for fixed income (both government bonds and interest rates) via both inflationary pressures and upward pressure on real interest rates.

Inflation: Inflation trends are captured using one-year changes in forecasts of CPI inflation. From 1990 onward forecast data is from Consensus Economics. Prior to 1990, we use one-year changes in realized year-on-year CPI inflation, lagged one quarter (this definition is equivalent to changes in forecasts assuming that CPI inflation follows a random walk). The series is from the OECD. Increasing inflation is assumed to be bearish for equities (see Katz and Lustig (2017)), bullish for currencies (see Clarida and Waldman (2008)), and bearish for fixed income.

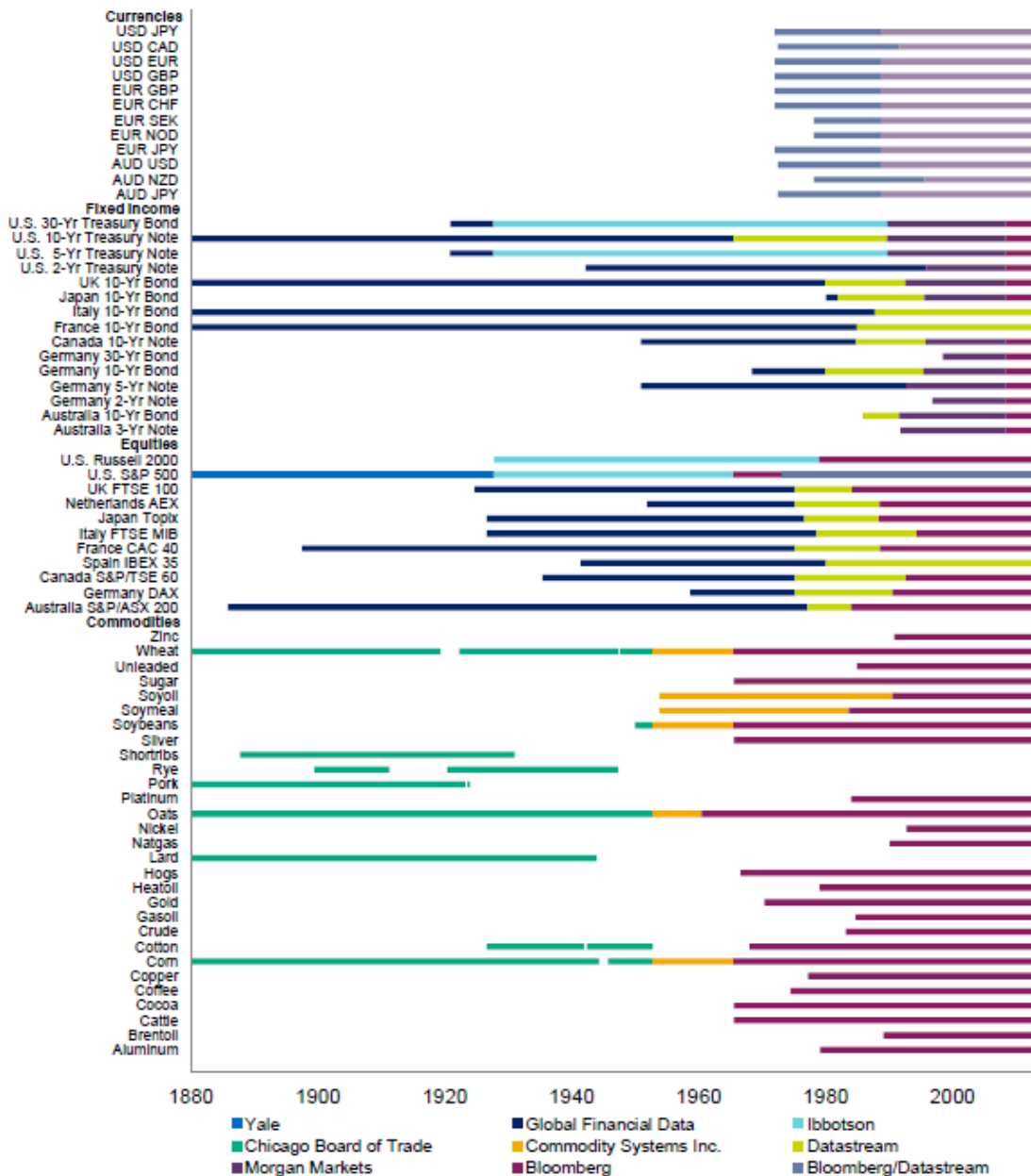
International Trade: International trade trends are captured using one-year changes in spot exchange rates against an export-weighted basket. Data is from DataStream. A depreciating currency is bullish for equities (exports become more competitive), bearish for currencies (very similar to price momentum), bearish for fixed income (other things equal, a depreciating currency reduces the pressure on a central bank to reduce interest rates), and bearish for commodities (depreciation of the currencies of commodity consumers means commodities, which are generally priced in USD, are effectively more expensive).

Monetary Policy: Monetary policy trends are captured using one-year changes in the front end of the yield curve. From 1992 onwards, I use two-year yields, while prior to 1992 I use Libor and its international equivalents. Both data series are from Bloomberg. Expansionary monetary policy is bullish for equities (see Bernanke and Kuttner (2005)), bullish for currencies (see Eichenbaum and Evans (1995)), bullish for commodities, and bearish for fixed income.

Risk Sentiment: Changes in risk sentiment are captured using one-year equity market excess returns. Data is from DataStream. Increasing risk sentiment is bullish for equities, commodities, and currencies, and bearish for fixed income.

The model employs relatively simple measures as they afford long data availability and are less susceptible to concerns about data mining. The strategy is therefore intended as a proof of concept, and can potentially be enhanced by employing additional and improved measures of economic trends. Hypothetical performance is net of estimated transaction costs and net of 1.25% annual management fee and 20% performance fee.

Hypothetical Price Trend-Following Strategy Universe



Hypothetical Economic Trend-Following Strategy Universe

Equity index return data is from Bloomberg. Start dates are the earliest available date of the series:

- 1970: Australia, Germany, Canada, Spain, France, Italy, Japan, Netherlands, U.K., U.S.
- 1975: Switzerland
- 1980: Denmark, Hong Kong, Sweden
- 1988: New Zealand

Government bond return data is from Bloomberg and DataStream. Start dates are

- 1970: Germany, Canada, U.K., U.S.
- 1980: Japan
- 1981: Switzerland
- 1985: Denmark
- 1986: Australia
- 1987: Sweden

Currency return data is from Citi and Reuters. Start dates are

- 1971: Germany, Japan, Switzerland, U.K.
- 1972: Australia, Canada

Interest rate futures return data is from IFS. Start dates are

- 1978: New Zealand, Sweden
- 1987: U.S.
- 1988: U.K.
- 1989: Australia, Europe (Euribor)
- 1991: Canada, New Zealand, Switzerland

Commodity futures return data is from Bloomberg. Start dates are

- 1970: Cattle, Corn, Cotton, Hogs, Soybeans, Soymeal, Soyoil, Sugar, Wheat
- 1974: Coffee
- 1979: Heat Oil
- 1983: Crude Oil
- 1984: Gas Oil
- 1985: Unleaded
- 1989: Brent Oil
- 1990: Natural Gas
- 1991: Zinc
- 1993: Nickel

The **Hypothetical Alternative Trend-Following Strategy** is an equal-weighted combination of 1-month, 3-month, and 12-month trend-following strategies for markets across 6 major asset groups – equity factor portfolios, credit indices, interest rate swaps, emerging currencies, alternative commodities, and volatility futures – from January 1990 onward. We construct the strategies using the largest number of assets for which return data exist at each point in time. We use futures returns when they are available. The strategy targets long-term volatility target of 10%. Returns are net of estimated transaction costs and 1.25% & 20% fees.

Disclosures

This document has been provided to you solely for information purposes and does not constitute an offer or solicitation of an offer or any advice or recommendation to purchase any securities or other financial instruments and may not be construed as such. The factual information set forth herein has been obtained or derived from sources believed by the author and AQR Capital Management, LLC ("AQR"), to be reliable, but it is not necessarily all-inclusive and is not guaranteed as to its accuracy and is not to be regarded as a representation or warranty, express or implied, as to the information's accuracy or completeness, nor should the attached information serve as the basis of any investment decision. This document is not to be reproduced or redistributed without the written consent of AQR. The information set forth herein has been provided to you as secondary information and should not be the primary source for any investment or allocation decision.

Past performance is not a reliable indicator of future performance.

This document is not intended to, and does not relate specifically to any investment strategy or product that AQR offers. It is being provided merely to provide a framework to assist in the implementation of an investor's own analysis and an investor's own view on the topic discussed herein. This presentation is not research and should not be treated as research. This presentation does not represent valuation judgments with respect to any financial instrument, issuer, security, or sector that may be described or referenced herein and does not represent a formal or official view of AQR.

HYPOTHETICAL PERFORMANCE RESULTS HAVE MANY INHERENT LIMITATIONS, SOME OF WHICH ARE DESCRIBED BELOW. NO REPRESENTATION IS BEING MADE THAT ANY ACCOUNT WILL OR IS LIKELY TO ACHIEVE PROFITS OR LOSSES SIMILAR TO THOSE SHOWN. IN FACT, THERE ARE FREQUENTLY SHARP DIFFERENCES BETWEEN HYPOTHETICAL PERFORMANCE RESULTS AND THE ACTUAL RESULTS SUBSEQUENTLY ACHIEVED BY ANY PARTICULAR TRADING PROGRAM. ONE OF THE LIMITATIONS OF HYPOTHETICAL PERFORMANCE RESULTS IS THAT THEY ARE GENERALLY PREPARED WITH THE BENEFIT OF HINDSIGHT. IN ADDITION, HYPOTHETICAL TRADING DOES NOT INVOLVE FINANCIAL RISK, AND NO HYPOTHETICAL TRADING RECORD CAN COMPLETELY ACCOUNT FOR THE IMPACT OF FINANCIAL RISK IN ACTUAL TRADING. FOR EXAMPLE, THE ABILITY TO WITHSTAND LOSSES OR TO ADHERE TO A PARTICULAR TRADING PROGRAM IN SPITE OF TRADING LOSSES ARE MATERIAL POINTS WHICH CAN ALSO ADVERSELY AFFECT ACTUAL TRADING RESULTS. THERE ARE NUMEROUS OTHER FACTORS RELATED TO THE MARKETS IN GENERAL OR TO THE IMPLEMENTATION OF ANY SPECIFIC TRADING PROGRAM WHICH CANNOT BE FULLY ACCOUNTED FOR IN THE PREPARATION OF HYPOTHETICAL PERFORMANCE RESULTS AND ALL OF WHICH CAN ADVERSELY AFFECT ACTUAL TRADING RESULTS.

The hypothetical performance results contained herein represent the application of the quantitative models as currently in effect on the date first written above and there can be no assurance that the models will remain the same in the future or that an application of the current models in the future will produce similar results because the relevant market and economic conditions that prevailed during the hypothetical performance period will not necessarily recur. Discounting factors may be applied to reduce suspected anomalies. This backtest's return, for this period, may vary depending on the date it is run. Hypothetical performance results are presented for illustrative purposes only. In addition, our transaction cost assumptions utilized in backtests, where noted, are based on AQR Capital Management, LLC's, ("AQR")'s historical realized transaction costs and market data. Certain of the assumptions have been made for modeling purposes and are unlikely to be realized. No representation or warranty is made as to the reasonableness of the assumptions made or that all assumptions used in achieving the returns have been stated or fully considered. Changes in the assumptions may have a material impact on the hypothetical returns presented. Actual advisory fees for products offering this strategy may vary. The information generated by the analysis regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results, and are not guarantees of future results. The analyses provided may include certain statements, estimates and targets prepared with respect to, among other things, Hypothetical and anticipated performance of certain assets. Such statements, estimates, and targets reflect various assumptions by AQR concerning anticipated results that are inherently subject to significant economic, competitive, and other uncertainties and contingencies and have been included solely for illustrative purposes.

The views expressed reflect the current views as of the date hereof, and neither the author nor AQR undertakes to advise you of any changes in the views expressed herein. It should not be assumed that the author or AQR will make investment recommendations in the future that are consistent with the views expressed herein, or use any or all of the techniques or methods of analysis described herein in managing client accounts. AQR and its affiliates may have positions (long or short) or engage in securities transactions that are not consistent with the information and views expressed in this presentation.

The information contained herein is only as current as of the date indicated and may be superseded by subsequent market events or for other reasons. Charts and graphs provided herein are for illustrative purposes only. The information in this presentation has been developed internally and/or obtained from sources believed to be reliable; however, neither AQR nor the author guarantees the accuracy, adequacy, or completeness of such information. Nothing contained herein constitutes investment, legal, tax, or other advice, nor is it to be relied on in making an investment or other decision.

There can be no assurance that an investment strategy will be successful. Historic market trends are not reliable indicators of actual future market behavior or future performance of any particular investment, which may differ materially, and should not be relied upon as such. Target allocations contained herein are subject to change. There is no assurance that the target allocations will be achieved, and actual allocations may be significantly different from those shown here. This presentation should not be viewed as a current or past recommendation or a solicitation of an offer to buy or sell any securities or to adopt any investment strategy.

The information in this presentation might contain projections or other forward-looking statements regarding future events, targets, forecasts, or expectations regarding the strategies described herein and is only current as of the date indicated. There is no assurance that such events or targets will be achieved and might be significantly different from that shown here. The information in this presentation, including statements concerning financial market trends, is based on current market conditions, which will fluctuate and may be superseded by subsequent market events or for other reasons. Performance of all cited indices is calculated on a total return basis with dividends reinvested.

The investment strategy and themes discussed herein may be unsuitable for investors depending on their specific investment objectives and financial situation. Please note that changes in the rate of exchange of a currency might affect the value, price, or income of an investment adversely. Neither AQR nor the author assumes any duty to, nor undertakes to update forward-looking statements. No representation or warranty, express or implied, is made or given by or on behalf of AQR, the author, or any other person as to the accuracy and completeness

or fairness of the information contained in this presentation, and no responsibility or liability is accepted for any such information. By accepting this presentation in its entirety, the recipient acknowledges its understanding and acceptance of the foregoing statement. Diversification does not eliminate the risk of experiencing investment losses.

"Expected" or "Target" returns or characteristics refer to expectations based on the application of mathematical principles to portfolio attributes and/or historical data, and do not represent a guarantee. These statements are based on certain assumptions and analyses made by AQR in light of its experience and perception of historical trends, current conditions, expected future developments and other factors it believes are appropriate in the circumstances, many of which are detailed herein. Changes in the assumptions may have a material impact on the information presented.

Broad-based securities indices are unmanaged and are not subject to fees and expenses typically associated with managed accounts or investment funds. Investments cannot be made directly in an index.

The MSCI USA Index is designed to measure the performance of the large and mid cap segments of the US market.

The MSCI World Index is a free float-adjusted market capitalization weighted index that is designed to measure the equity market performance of developed markets.

The MSCI EAFE Net TR Unhedged Index measures the performance of large- and mid-cap equities across 21 developed markets outside the United States and Canada.

The MSCI Emerging Markets Index is a free float-adjusted market capitalization index that is designed to measure equity market performance of emerging markets.

The Russell 2000 Index is a small-cap U.S. stock market index that makes up the smallest 2,000 stocks in the Russell 3000 Index.

The Bloomberg Global Aggregate Bond Index is a flagship measure of global investment grade debt from twenty-four local currency markets. This multi-currency benchmark includes treasury, government-related, corporate and securitized fixed-rate bonds from both developed and emerging markets issuers.

The Bloomberg Global Treasury Index tracks fixed-rate local currency government debt of investment grade countries.

The Bloomberg Global High Yield Index is a multi-currency flagship measure of the global high yield debt market. The index represents the union of the US High Yield, the Pan-European High Yield, and Emerging Markets (EM) Hard Currency High Yield Indices.

The Bloomberg US Corporate Bond Index measures the return of the investment-grade, fixed-rate, taxable corporate bond market.

The Bloomberg US Corporate Bond Credit Excess Index measures the return of the investment-grade, fixed-rate, taxable corporate bond market in excess of duration-matched US Treasuries.

The Bloomberg US Treasury Inflation-Linked Bond Index tracks the performance of inflation-protected public obligations issued by the U.S. Treasury, commonly known as TIPS.

The SG Trend Index is designed to track the largest 10 (by AUM) CTAs and be representative of the managed futures trend-following space. The index uses equal-weighted net daily returns and is reconstituted annually. The AQR Funds - AQR Managed Futures Strategy Fund is a constituent of the SG Trend Index.

The Cambridge Associates U.S. Private Equity Index is calculated based on data compiled from approximately 2,000 funds, including fully liquidated partnerships.

The Bloomberg Commodity Index is made up of 23 exchange-traded futures on physical commodities, representing 21 commodities which are weighted to account for economic significance and market liquidity.

The Cboe S&P 500 5% Put Protection Index (PPUT) tracks the value of a hypothetical portfolio of securities (PPUT portfolio) designed to protect an investor from negative S&P 500 returns. The PPUT portfolio is composed of S&P 500® stocks and of a long position in a one-month 5% out-of-the-money put option on the S&P 500 (SPX put).

The FTSE NAREIT All REITs Index is a market capitalization-weighted index that includes all tax-qualified U.S. REITs.

The NCREIF Property Index measures the performance of real estate investments on a quarterly basis and evaluates the rate of returns in the market. The NPI covers properties that are acquired in place of institutional investors that are exempted from taxes in the fiduciary environment.

The HFRI Equity Hedge Index is an equal-weighted index of funds maintaining positions both long and short in primarily equity and equity derivative securities.

The HFRI Macro Index is an equal-weighted index of funds which execute a broad range of strategies in which the investment process is predicated on movements in underlying economic variables and the impact these have on equity, fixed income, currency and commodity markets.

The HFRI Relative Value Index is an equal-weighted index of funds who maintain positions in which the investment thesis is predicated on realization of a valuation discrepancy in the relationship between multiple securities.

Gross performance results do not reflect the deduction of investment advisory fees and other expenses, which would reduce an investor's actual return.

There is a risk of substantial loss associated with trading commodities, futures, options, derivatives, and other financial instruments. Before trading, investors should carefully consider their financial position and risk tolerance to determine whether the proposed trading style is appropriate. Investors should realize that when trading futures, commodities, options, derivatives, and other financial instruments, one could lose the full balance of their account. It is also possible to lose more than the initial deposit when trading derivatives or using leverage. All funds committed to such a trading strategy should be purely risk capital.

Regional Disclosures

Australia: AQR Capital Management, LLC, is exempt from the requirement to hold an Australian Financial Services License under the Corporations Act 2001, pursuant to ASIC Class Order 03/1100 as continued by ASIC Legislative Instrument 2016/396 (as extended by amendment). AQR is regulated by the Securities and Exchange Commission ("SEC") under United States of America laws and those laws may differ from Australian laws.

Canada: This material is being provided to you by AQR Capital Management, LLC, which provides investment advisory and management services in reliance on exemptions from adviser registration requirements to Canadian residents who qualify as "permitted clients" under applicable Canadian securities laws. No securities commission or similar authority in Canada has reviewed this presentation or has in any way passed upon the merits of any securities referenced in this presentation and any representation to the contrary is an offence.

Dubai: AQR Capital Management (Europe) LLP (DIFC Representative Office) is regulated by the Dubai Financial Services Authority of the Dubai International Financial Centre as a Representative Office (firm reference number: F007651). Its principal place of business is Gate Village 10, Level 3, Unit 4, DIFC, Dubai, UAE. This marketing communication is distributed on behalf of AQR Capital Management, LLC.

UK: The information set forth herein has been prepared and issued by AQR Capital Management (Europe), LLP, a UK limited liability partnership with its office at 15 Bedford St, Covent Garden, London, WC2E 9HE, which is authorized and regulated by the UK Financial Conduct Authority ("FCA").

EU: AQR in the European Economic Area is AQR Capital Management (Germany) GmbH, a German limited liability company (Gesellschaft mit beschränkter Haftung; "GmbH"), with registered offices at Maximilianstrasse 13, 80539 Munich, authorized and regulated by the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), with offices at Marie-Curie-Str. 24-28, 60439, Frankfurt am Main und Graurheindorfer Str. 108, 53117 Bonn, to provide the services of investment advice (Anlageberatung) and investment broking (Anlagevermittlung) pursuant to the German Securities Institutions Act (Wertpapierinstitutsgesetz; "WpIG"). The Complaint Handling Procedure for clients and prospective clients of AQR in the European Economic Area can be found here: <https://ucits.aqr.com/Legal-and-Regulatory>.

AQR Capital Management (Asia): This presentation may not be copied, reproduced, republished, posted, transmitted, disclosed, distributed or disseminated, in whole or in part, in any way without the prior written consent of AQR Capital Management (Asia) Limited (together with its affiliates, "AQR") or as required by applicable law. This presentation and the information contained herein are for educational and informational purposes only and do not constitute and should not be construed as an offering of advisory services or as an invitation, inducement or offer to sell or solicitation of an offer to buy any securities, related financial instruments or financial products in any jurisdiction. Investments described herein will involve significant risk factors which will be set out in the offering documents for such investments and are not described in this presentation. The information in this presentation is general only and you should refer to the final private information memorandum for complete information. To the extent of any conflict between this presentation and the private information memorandum, the private information memorandum shall prevail. The contents of this presentation have not been reviewed by any regulatory authority in Hong Kong. You are advised to exercise caution and if you are in any doubt about any of the contents of this presentation, you should obtain independent professional advice.

AQR Capital Management (Asia) Limited is licensed by the Securities and Futures Commission ("SFC") in the Hong Kong Special Administrative Region of the People's Republic of China ("Hong Kong") pursuant to the Securities and Futures Ordinance (Cap 571) (CE no: BHD676). AQR Capital Management (Asia) Limited, Unit 2023, 20/F, One IFC, 1 Harbour View Street, Central Hong Kong, Hong Kong. Licensed and regulated by the Securities and Futures Commission of Hong Kong (CE no: BHD676).

China: This document does not constitute a public offer of any fund which AQR Capital Management, LLC ("AQR") manages, whether by sale or subscription, in the People's Republic of China (the "PRC"). Any fund that this document may relate to is not being offered or sold directly or indirectly in the PRC to or for the benefit of, legal or natural persons of the PRC.

Further, no legal or natural persons of the PRC may directly or indirectly purchase any shares/units of any AQR managed fund without obtaining all prior PRC's governmental approvals that are required, whether statutorily or otherwise. Persons who come into possession of this document are required by the issuer and its representatives to observe these restrictions.

Singapore: This document does not constitute an offer of any fund which AQR Capital Management, LLC ("AQR") manages. Any fund that this document may relate to and any fund related prospectus that this document may relate to has not been registered as a prospectus with the Monetary Authority of Singapore. Accordingly, this document and any other document or material in connection with the offer or sale, or invitation for subscription or purchase, of shares may not be circulated or distributed, nor may the shares be offered or sold, or be made the subject of an invitation for subscription or purchase, whether directly or indirectly, to persons in Singapore other than (i) to an institutional investor pursuant to Section 304 of the Securities and Futures Act, Chapter 289 of Singapore (the "SFA") or (ii) otherwise pursuant to, and in accordance with the conditions of, any other applicable provision of the SFA.

Korea: Neither AQR Capital Management (Asia) Limited or AQR Capital Management, LLC (collectively "AQR") is making any representation with respect to the eligibility of any recipients of this document to acquire any interest in a related AQR fund under the laws of Korea, including but without limitation the Foreign Exchange Transaction Act and Regulations thereunder. Any related AQR fund has not been registered under the Financial Investment Services and Capital Markets Act of Korea, and any related fund may not be offered, sold or delivered, or offered or sold to any person for re-offering or resale, directly or indirectly, in Korea or to any resident of Korea except pursuant to applicable laws and regulations of Korea.

Japan: This document does not constitute an offer of any fund which AQR Capital Management, LLC ("AQR") manages. Any fund that this document may relate to has not been and will not be registered pursuant to Article 4, Paragraph 1 of the Financial Instruments and Exchange Law of Japan (Law no. 25 of 1948, as amended) and, accordingly, none of the fund shares nor any interest therein may be offered or sold, directly or indirectly, in Japan or to, or for the benefit, of any Japanese person or to others for re-offering or resale, directly or indirectly, in Japan or to any Japanese person except under circumstances which will result in compliance with all applicable laws, regulations and guidelines promulgated by the relevant Japanese governmental and regulatory authorities and in effect at the relevant time. For this purpose, a "Japanese person" means any person resident in Japan, including any corporation or other entity organised under the laws of Japan.

Request ID: 406063



aqr.com